

# Universe24: a cubic lattice of integers stepped by one rule and read through clicks

Alon Gonen

Independent researcher, Haifa, Israel

alon@defounder.ai ORCID 0009-0008-2701-4599

October 2026

## Abstract

We present one method of computing physics: a cubic lattice whose Nodes hold whole numbers, stepped by one integer recurrence and read only through declared detectors' clicks. Measurement is the meeting of the record run forward with the one read backward from the click: the detector draws, the lattice draws nothing, and the one non-local element is the credit's one draw through the common root. The three computed worlds are derivations: the two-slit pattern is the line stepped on its world, and Bell's  $S = 2\sqrt{2}$  and GHZ's  $M = -4$  follow, for a pair laid alike as one event, from the line's linearity under one declared credit. On it the classical wave equation, the quantum of action and the metric of gravity are three readings of one line, with nothing continuous beneath. The recurrence steps every Node from its present, its past and six neighbours by one integer division, reversibly and covariantly, conserving a quadratic form. Newton's gravitation is reflected,  $G$  as a reading; Einstein's weak field is reflected in the clock's potential under one named assumption, a Link clocked twice by its Node, fitted to two of its clicks: the exponential metric,  $\beta = \gamma = 1$  in that potential as the gap closes, a shadow 4.6 percent above Schwarzschild's; against Kepler's potential the computed pair's bending and redshift depart by 25 and 33 percent, closing below  $10^{-19}$ . The paper claims the method and not nature: every result is a supporting finding, offered for the reader's test, and what the method does not yet give is listed.

**Keywords:** integer lattice rule; reversible cellular automaton; measurement as a meeting of past and future; Bell's inequality; weak-field gravity; limiting speed on a lattice

## 1 Introduction

**The question.** The object of this paper is one method of computing physics: a GameBoard of Nodes stepped by one local, reversible rule in whole numbers, read only through clicks, the one measurement there is. On it the classical wave equation, the quantum of action and the metric of gravity are three readings of one line, Rule3: the line itself is the lattice wave equation, its count read in quanta is the click, and its paces read as a clock and a ruler are the metric, with nothing continuous beneath. The question is whether that method can give the clicks that nature shows, and with how little state, non-locality and randomness; the fifteen formulas and the three computed worlds below are examples of what the method yields, not its scope, and Section 10.3 lists what it does not yet give. The paper's one claim is the method: a GameBoard of integers stepped by one rule and read into the world through clicks, from which the forms of nature are derived in the order stated. It does not claim that nature is such a lattice; every derivation, comparison and prediction below is a supporting finding for the method, each with its status, and none is the paper's thesis. It starts from one statement and never contradicts it (ASSUMPTION): the world is the meeting of the past with the future; a click is that meeting, the one measurement there is; and an observer sees only clicks (the words are defined at the end of this section). Every physical number of the paper is a reading of one integer rule with its declared integers, the pairs,  $\Gamma$ ,  $T$ , the two holders' weights, gravity's  $E_g$  and the sign holder's  $k$ , and the vacuum content (Section 6.6); of nature's other clicks it gives the forms of Section 8 at the computed pair, with the findings of Section 10.3 against them. A Node holds a record's present and its past, two integer levels, and the rule makes the record's future from the two and from the six neighbouring Nodes. The rule is covariant under the lattice's translations and under the symmetry group of the cube, of order 48; no boost is among them (Section 3.1). The lattice holds nothing but events that split: at every Node a level splits by the rule to the six neighbours. Run forward from an emitter, the split events are the future, every path at once; read backward from a realised click by the same rule, they are the past, and through two slits both pass both gaps. A click is where the two meet at a declared detector, and which meeting is realised is the one thing the model does not determine, the one non-local

element of the description (Section 5.2). What appears as a wave is the future, what appears as a particle is the present, the click, and the track is the past (Section 5.1). No theorem of minimality is claimed.

**The rule in one line.** Everything in the paper follows from one line of integer arithmetic, Rule3, together with one write and three readings. At every Node, in every interval and for every family of physical values [1, the line; the booking; the primitives]:

|                |                                                                                                                                                                                                                                                                                                                                              |     |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Rule3:         | $w a_{\text{next}} + r' = \sum_{j \sim i} R_{ij} \text{arr}_j + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{den} \Gamma^2, \quad R_{ij} = 2 \text{num} p_a(i, j)^2 = 2 \text{num} p_i^2 q_{ij}^2 / \Gamma^2,$ $S = 12 \text{den} \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - \sum_{j \sim i} R_{ij};$ |     |
| the one write: | $a \text{ held part at a Node gains } \left( \sum_f w_f q_f(\text{Node}) + r \right) \text{div } E, \text{ the remainder } r \text{ kept there;}$                                                                                                                                                                                            | (1) |
| the readings:  | $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i, \quad W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}},$ $F_{ij} = \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j).$                                                                                |     |

In the first line  $a_{\text{now}}$ ,  $a_{\text{before}}$  and  $a_{\text{next}}$  are the family's level at the Node now, one interval before and one interval later: the present, the past and the future.  $\text{arr}_j$  is the arrival through the Port to the neighbour  $j$ , that neighbour's level, the six Ports summed.  $r$  and  $r'$  are the remainder kept at the Node before and after the step,  $\Gamma$  the Node clock.  $[\text{num}, \text{den}]$  is the family's pair, and  $\cos \omega_0 = \text{num}/\text{den}$  is its rotation per interval at wave number zero.  $p_0$  is the Node's clock pace,  $p_i$  its own pace and  $p_a(i, j)$  the pace of the Link through the Port, the Node's pace times the Link's factor  $q_{ij}/\Gamma$ , the factor read the same from both ends and the pace the reading Node's own. All are the Node clock lowered by the field levels the family reads there (Section 3.4). At the vacuum's paces the two Links of an axis share one  $R_a = 2 \text{num} \Gamma^2$  and  $S = 0$ .  $\text{div}$  is the floor division. The second line is the only way a field's level is ever written: the sum over the sourcing families  $f$  of each family's own quantity  $q_f$  at the Node, weighted by  $w_f$ , plus the part's one remainder, divided by the part's wall  $E$ . The third line is what is read from a record and never written: its form  $D_i$ , which sources the fields; its Wronskian  $W_i$ , its charge; and its current  $F_{ij}$  into the Node  $i$  from its neighbour  $j$ , which a detector reads through its boundary in quanta of  $W_c = 3 \text{den} T$ , with  $T$  the quantum's norm, the normalisation of the quadratic form.

**The formulas derived.** From Eq. (1), in the paper's order, each with its mark in Table 1 and its algebra in the supplement: light's band, its dispersion and its index in a well (Sections 3.3 and 3.4); the lattice Klein-Gordon equation and its Schrödinger limit (Section 3.3); the conserved form and the share identity (Sections 4.3 and 4.4); the Wronskian as the charge and the holders' rests (Sections 5.5 and 6.4); the bound body's functional and its stability (Section 7.2); the adiabatic invariant and the count's drift (Section 7.4); the clock's rate, the bending, Shapiro's delay, the perihelion and the shadow's capture radius, the moving clock, de Broglie's and Planck's relations and  $E = mc^2$  in the band's form, the content's index for every family and the fall as refraction (Section 8); Bell's  $E$  and  $S$ , Tsirelson's bound and GHZ's  $M$  under the declared credit (Section 9.4); the fine-structure constant as a coefficient of the file, the hierarchy of the two forces between families, the units and the highest photon energy, and Newton's  $G$  as a reading (Section 10.1). Fifteen formulas, several with their companions, from five fields, every one an algebraic consequence of the line and of the inputs named beside it.

**What the paper contributes.** We state the rule and what is exact about it, proved in the supplement, reversibility to the bit, the conserved form with its current, the momentum flux as the wave's own stress (Section 4). We formulate measurement as the meeting of the forward and the backward record at a declared detector, the one non-local element named (Section 5). We obtain the families as bands of the rule and choose a universe with as few free numbers as the rule allows (Section 6). We show that bodies are fixed points, prove their stability under the dilation and state how they cluster (Section 7). We compute the weak-field forms of gravity and the moving clock's factor, each marked by what it rests on, in the model's own potential and in the one nature's tests read (Section 8). And we state the two slits and Bell as derivations of what each must give, neither experiment performed here (Section 9).

**What stands, and on what.** Five things, each with its mark. (i) The core: Rule3 and the boxed lines that follow it, the paces, the conserved form, the share, the Wronskian, Kepler's potential and  $G$  as a reading, are theorems of the line or derived from it with no number of nature in them, and the step is reversible to the bit (Section 4); what the checks of this paper's closing corrected lay beside them, in secondary derivations, marks and digits, and not in the line. (ii) One rule in integers reaches Einstein's weak field: from the composed paces alone, under the one named assumption of Section 3.4, come the exponential metric, the bending  $4U_b$ , Shapiro's delay, the redshift and the push  $E(1 + \beta^2)\nabla U$ , with  $\beta = \gamma = 1$  in the clock's potential, the assumption fitted by two of those clicks and the others its predictions, and the forms against Kepler's potential closing on them as the gap closes (Section 8). (iii) The click is

the one measurement, and the non-locality is accounted for in one place: Bell's 478/169 and the GHZ's  $-4$  follow from the declared credit, the two slits' 278 stands against the blind 273, and the one non-local element is the credit's one draw through the root; the meeting of the past with the future at the detector is one picture (Section 5). (iv) Every claim carries its mark and its fence, and what fails is named with its number,  $\alpha_1 = -8$ , hydrogen's  $-0.043 R$ , the shelved ion's telegraph (Section 10.3); nothing is claimed beyond its mark. (v) Everything is recomputable: the files, the engine, the scripts and the documents are public (Section 3.7), and every number that stands in the paper was recomputed twice, independently, from them. All of it is derived from one chosen set of rules, of which the engine fixes only the recurrence's form: other families, primitives and integers can be taken and other things built (Section 3.7).

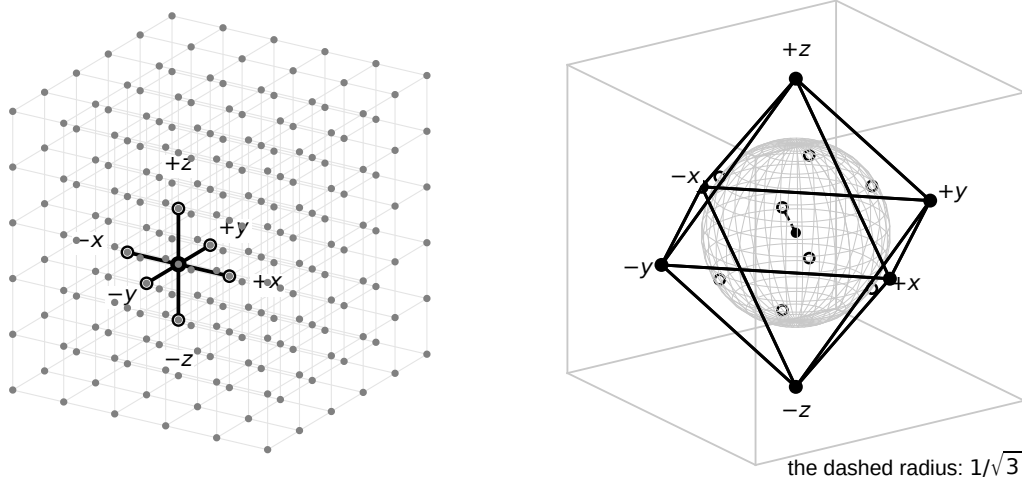
**What is put in, and the three kinds of claim.** The paper's claims are of three kinds, and the text keeps them apart. *Found*: what is proved or derived from the rule with no input chosen for it, the fifteen formulas above and the click formulas of Table 2 marked derived; the findings against nature are comparisons, listed in Section 10.3. *Assumed, chosen or declared*: the assumption and the credit below, the universe's rows, the detector's four declarations, its unit of credit among them, the electron as the free quantum of matter and the matter pair's denominator as the family's own integer (Section 10.1). *Held to be computable and not yet computed*: the bodies at the closing gap, the moving fixed point of a body and the absorbed fraction with the body's write; the explicitly named hypotheses are stated where they stand, among them light's gravity entered as a derived hypothesis (Section 10.2), the carrier that would mend the lattice's frame and the nuclear holders (Section 10.3). One assumption is named with the forms, that a Link is clocked twice by its reading Node's own clock,  $hN = 1$ , fixed by the bending and the perihelion (Section 3.4); the forms against Kepler's potential reach Einstein's only as the matter family's gap closes (Section 8). One choice was made with a known result in view, and we state it as an input, not a prediction: a click's share is the square of the meeting, a credit chosen to reach nature's correlations, so Bell's  $2\sqrt{2}$  follows from it and does not test it. The file's numbers are nature's and not the rule's: the vacuum content, a declaration; the vacuum write's rate,  $\Lambda$ , the one declared number of the hypothesis of Section 7.5; the binding range, a world's choice; gravity's weight  $E_g$  and the sign holder's write weight  $k$ , read from  $G$  and from  $\alpha$ ; and the gaps, read from the masses (Section 10.2, Table 4). Table 1 lists every result of the paper with its kind, its fence and its inputs. What is not claimed: that nature is a lattice; that any number of the computed pair [2, 3] is nature's; that a comparison with nature is more than a consistency statement at the stated order; or that a formula known in the continuum is new where the paper gives its lattice form, the known form standing beside it in every row. Every derivation was checked twice, independently, within the project and, where a world exists, by the implementation's checks, not by an independent referee. Where the text says exact it means exact in the rational arithmetic before the rounding, with the integer remainders bounded as stated. And a line decided after a derivation is stated as the law with the implementation's run pending, a run that differed from its prediction being a defect of the implementation to repair and never a result. The paper computes one world of the many the method admits: every number of the files and every declared row, a family, a holder, a read, an instrument, can be changed and the same rule run again; what is brought here is one plane of them, with every derivation and every new formula shown, and the claim is only that it approaches nature's clicks in the order stated, never that nature is so.

**How a world is built and tested.** The method's statement, with its pointers. A world file declares, and nothing else: a family's pair and dimension; a holder's sources, divisor and rest; the bodies and the messages; a detector's four declarations, region, window, floor and whole, and its seed; the faces. The engine holds no number, no formula, no family name and no flag, and every primitive is one folder found by its name [2, how to run a world; how to add a family]. One change, from numbers the paper keeps: the matter pair [2, 3] against [999, 1000] moves  $c_m/c$  from 0.867 to 1 and the computed pair's  $\gamma_K$  and  $\beta_K$  from 1.5 and 1.25 to 1 (Section 8); a binding holder's range  $R$  turns a declared body from unbound to bound (Section 6.4). The procedure, the two slits the example: the blind row is written from the law's lines before the run, the run is made, and the gate reports MATCH or MISS (Sections 3.7 and 9). Three worlds are run and gated in this paper, the two slits, Bell's gate and the GHZ gate; every other experiment the paper sets against nature is derived from the law's lines and not run, each with its mark and fence where it stands: Einstein's weak-field forms, the bending, Shapiro's delay, the redshift as the clock at two heights and the perihelion at the small gap (S.15, S.19, S.20), the shadow (S.30), the push on a moving record (S.58), the two slits' own blind row written before the run (Section 9.2), the moving clock (S.21, S.22), the magnetism of moving charges (S.33), the Eötvös ratio (S.47), the lens's one-quantum image (Eq. (17)), the clusters' deceleration (S.29), the atom (S.46), Rabi's absorption line (S.43), the gravity of light (Section 10.2), the anticoincidence (S.57), the Zeno effect (S.59), the preferred frame's  $\alpha_1$  and  $\alpha_2$  against Gravity Probe B and the Sun's spin (S.35), derived and failing, and, derived under a declaration, the shelved ion's telegraph and the excited body's giving (the rate declared; Section 10.3) and the nuclear binding (S.45, the weight declared); their engine worlds are the gate after the freeze. Rule3 is fixed and no rule is added: a new line of physics enters as a hypothesis under its own name after the three tests, generic, vector and local, which is how every family and holder of this paper entered (Section 3.6).

**Table 1** The results of the paper, each with its kind, its fence and what it rests on. A calibration is a number taken from nature; EXPERIMENT a number the model does not fix.

| Result                                                                                                                                                                                                                                           | Kind                                                         | Fence             | Rests on                                                                           | Section   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------|-----------|
| Rule3's step is reversible to the bit; the interval under the condition of Section 4.1                                                                                                                                                           | theorem                                                      | GameBoard         | Rule3                                                                              | 4.1       |
| The gate says MATCH on the two slits', Bell's and GHZ's worlds: $N = 278$ , $S = 478/169$ , $M = -4$                                                                                                                                             | computed, a run                                              | clicks            | the implementation                                                                 | 4.1       |
| The rule's acts form no product of levels                                                                                                                                                                                                        | theorem                                                      | GameBoard         | Rule3                                                                              | 4.2       |
| A conserved form, its current, and the momentum flux as the wave's stress                                                                                                                                                                        | theorem                                                      | GameBoard         | Rule3                                                                              | 4.3       |
| The count is the record's share; the total share is never negative                                                                                                                                                                               | theorem                                                      | GameBoard         | Rule3                                                                              | 4.4       |
| Three exact massive rotations                                                                                                                                                                                                                    | theorem                                                      | GameBoard         | Rule3                                                                              | 6.1       |
| Light's speed $1/\sqrt{3}$ , its dispersion, and no dispersion along the cube's diagonal                                                                                                                                                         | derived                                                      | GameBoard         | Rule3 for the pair $[1, 1]$                                                        | 3.3       |
| A click is the meeting of two quanta, decided in the credit, written at one Node; no signal                                                                                                                                                      | assumption, derived                                          | clicks            | the meeting, the detector's draw and write                                         | 5         |
| A field's reach from its gap, Yukawa's form                                                                                                                                                                                                      | derived                                                      | GameBoard         | Rule3 for a gapped pair                                                            | 6.4       |
| The stability of a body under the dilation, to first order in the well                                                                                                                                                                           | theorem                                                      | GameBoard         | the body's functional                                                              | 7.2       |
| A body's clicks: no light from a steady rotation, light at a breathing body's beat, the resonance at $2\omega_b$ , the count's drift                                                                                                             | derived                                                      | clicks            | Rule3 at the body's Nodes                                                          | 7.4       |
| A cluster: the three levels at a Node as the nesting, three kinds of binding, the click the only thing between clusters, the hierarchy recursive                                                                                                 | assumption, derived                                          | GameBoard, clicks | the universe is bound; Rule3                                                       | 7.5       |
| The gravitational redshift, $z = f(\omega_0) U$                                                                                                                                                                                                  | derived                                                      | clicks            | the clock's read                                                                   | 8         |
| The bending and the perihelion, which fixed the one assumption; the delay, the redshift's factor and the shadow's capture radius $2e m$ , its predictions                                                                                        | derived under the one assumption, fitted by the first two    | clicks            | a Link clocked twice by its Node's clock, $hN = 1$                                 | 8         |
| The same forms against Kepler's potential: $\gamma_K$ , $\beta_K$ and $\alpha_K$ from the pair alone, closing with the gap                                                                                                                       | derived                                                      | clicks            | the fall's formula                                                                 | 8         |
| The moving clock: its fourth order and its kinetic scale $c_m$ , Lorentz's as the gap closes                                                                                                                                                     | derived                                                      | clicks            | Rule3's band                                                                       | 8         |
| The force between moving charges, Coulomb's over $\gamma$                                                                                                                                                                                        | derived; the implementation's run a check of its bookkeeping | clicks            | the rule's continuity identity, the sign's covector                                | 8         |
| De Broglie, the fall, Newton's pull; the content's index for every family, the lens                                                                                                                                                              | derived                                                      | GameBoard, clicks | the band and the field's rest                                                      | 8         |
| Newton's pull, the clock's $G$ and Kepler's $G$                                                                                                                                                                                                  | derived                                                      | GameBoard, clicks | Rule3 for $[1, 1]$ and $E_g$                                                       | 10.1      |
| $\alpha_{1aw} = (3\sqrt{3}/8\pi) k/(\Gamma E_s)$ between two quanta, $k$ the sign holder's declared weight; charge universality; the two forces' ratio between families $(m_p/m_e)^2$ ; the units; the share's ceiling; the gap below $10^{-19}$ | derived; $k$ declared                                        | clicks            | the rotation, Poisson's rest, de Broglie; the quantum of a family; LHAASO          | 10.1      |
| The two-slit fringe row and the lattice's arrival interval                                                                                                                                                                                       | computed                                                     | clicks            | Rule3 on the declared world                                                        | 9.2       |
| Bell's $E = \cos 2(a - b)$ , $S = 2\sqrt{2}$ , $S(\rho)$ and Tsirelson's $2\sqrt{1 + \rho^2}$ ; every local credit at most 2 (theorem)                                                                                                           | fitted; derived                                              | clicks            | the credit rule, the meeting's square                                              | 9.4       |
| The vacuum content 0; gravity's events at light's speed in every level                                                                                                                                                                           | assumption; implemented                                      | clicks            | declared; every row reads the content                                              | 6.5       |
| The binding range, 20 Links                                                                                                                                                                                                                      | a world's choice                                             | GameBoard         | declared, a reading of an earlier round the families replace nature's measurements | 6.4       |
| Gravity's weight $E_g$ and the sign holder's $k$ , the matter pair's gap, the Link's length                                                                                                                                                      | calibration, experiment                                      | clicks            |                                                                                    | 6.6, 10.2 |
| The two clocks shift alike; the findings against nature of Section 10.3                                                                                                                                                                          | derived; comparisons                                         | clicks            | the principle; Rule3                                                               | 10.3      |

**The three layers of the method.** The method has three layers, each with a fixed part and a free part. The board's rules (Section 3): which families and of what dimension, which holders, who reads what and how; Rule3 with its reads is fixed, the file's rows are free. The numbers (Section 6):  $\Gamma$ , the pairs, the holders' weights, gravity's  $E_g$  and the sign holder's  $k$ ,  $T$ , the vacuum content, the ranges, all free and as few as the rule allows. The click (Sections 5,



**Fig. 1** (a) The GameBoard in space: a cubic array of Nodes, here  $6 \times 6 \times 6$ , each joined to its six neighbours by Links through the Ports  $\pm x$ ,  $\pm y$ ,  $\pm z$  (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it. A Node keeps each family's two levels and its remainders there, and nothing else. (b) The Nodes one interval away from a Node form the octahedron  $|x| + |y| + |z| \leq 1$ , whose six vertices are the six Ports; its symmetry group is the cube's,  $O_h$ , the 24 rotations and, with the reflections, the 48 signed permutations of the axes, the group under which Rule3 is covariant (Section 3.1); its inscribed sphere, of radius  $1/\sqrt{3}$ , touches the eight faces on the cube's diagonals, and that radius is light's speed in Links per interval (Section 3.3). Both drawn from the definitions; no run.

5.3 and 7.4): how the board becomes a click, what the detector declares, what is credited and how a board mass is read as a click mass. Fixed here, common to every world, is the click's one form: the meeting, the bilinear credit, the draw by the shares, the detector's one write at the Node of the exchange and the postulate that only a click is a measurement; free, the world's, are the detector's four declarations, the unit it counts in among them.

**How to read the paper.** Every claim carries one of seven marks. **THEOREM:** proved for every lattice. **DERIVED:** obtained from theorems by argument. **COMPUTED:** a number obtained from the rule's lines before any run. **ASSUMPTION:** a line held by decision and named where it is used. **HYPOTHESIS:** a line outside the model, under its own name. **EXPERIMENT:** a number the model does not fix, with the experiment named that would fix it; the paper performs none. **FITTED:** an input chosen with a known result in view. The tables' status columns use the same marks, with calibration (a number taken from nature), declaration (a world's own number) and implemented (checked by the implementation) beside them. Every formula carries, beside its mark, its fence. **GAMEBOARD:** a reading of the lattice's own numbers that no instrument reports, a diagnostic. **CLICKS:** a formula of what a detector reports, the share  $Q_R/W_c$  on the forward orbit (Section 5.2) or a ratio of two such, the only kind that nature can be set beside. The algebraic steps of every derivation that the text states without its proof are given in the supplementary material (Online Resource 1), Derivations S.1 to S.60, numbered as added, each naming its section, cited as (S.*n*) where used. The Statements and Declarations at the end say how the text was worked out with a large language model under the author's direction, every statement reviewed by the author. The paper cites three documents of its archive by the names of their sections: the law [1], the engine [2] and the decisions [3].

**The words.** The physical location is a *Node*, and its complete local information is its *NodeState*. Nodes are joined by *Links* through six directional *Ports* per Node, and the lattice of Nodes is the *GameBoard* (Fig. 1(a)). A local change is an *Event*, and there is one local rule, Rule3. A *family* is one kind of physical value with its pair and what it holds. Its *record* is its levels over the GameBoard, two per Node with a remainder. A family of quanta has a *dimension*, 1 for one real line and 2 for a plane of two lines. A *body* is the set of Nodes on which a family's record stands bound, with no law of its own; a *cluster* is bodies in one another's wells; a *message* is a travelling record laid by the world. A *detector* is a region of Nodes declared in the file to report, and its *click* is one quantum credited to it, the one measurement; an *interval* is one step of the whole GameBoard; a family that holds a value for others to read is a *holder*. The words in full are S.49 [1, the objects][2, the words].

## 2 Related work

**Reversible cellular automata and digital physics.** The idea that the physical world is a discrete computation goes back to Zuse [4]; it was developed as reversible cellular automata by Fredkin, Toffoli and Margolus [5][6][7],

as a programme by Wolfram [8], and as a deterministic foundation beneath quantum theory by 't Hooft [9][10]. Quantum cellular automata put a unitary step on a lattice, from the lattice Weyl, Dirac and Maxwell equations [11] and quantum lattice gases [12] to the structure theorems of the reviews [13][14] and the derivation of Weyl's and Dirac's equations from an automaton's symmetries [15]; the present rule's step is an integer recurrence on two levels, not a unitary on amplitudes. Rule3 is a second-order cellular automaton in the sense of Toffoli and Margolus,  $a_{\text{next}} = f(a_{\text{now}}) - a_{\text{before}}$  [7], with an integer alphabet and one Euclidean division whose remainder stays at the Node. Feynman noted that a local classical computer reproduces no correlations beyond Bell's bound [16]; the model agrees, and places the excess in the declared credit of the click rather than in the lattice (Section 9.4).

**Time-symmetric accounts of measurement.** Wheeler and Feynman described radiation as an interaction between the source and the absorber, with advanced and retarded fields together [17]; Cramer's transactional interpretation has an offer wave forward and a confirmation wave backward, meeting at the absorber [18]. The two-state vector formalism evolves one state forward from the preparation and one backward from the outcome [19][20]; Price argued that a time-symmetric account is what Bell's theorem leaves open [21], and Wharton and Argaman showed that every locally mediated reformulation of quantum theory must let some mediating parameters depend on measurement settings in their future [22]. The model belongs to this class, the backward reading from a realised click depending on the detector's setting alone; it differs in that both directions are one integer line, exact to the bit, that their product is a reading and not a wave, and that the choice among the meetings is the one non-local element; the record is an ontic state in the terms of [23].

**Bell's theorem and local models.** Bell's inequality bounds every local account of a pair's correlations [24], in the form that experiments test [25], and a local model, a joint distribution and the inequalities are one condition [26]. The model's local credits give  $S = \sqrt{2}$  and  $S = 2$ , as they must (Section 9.4); its credit by the contrast reaches the quantum correlation only by discarding pairs, Pearle's road [27], below the efficiency bounds of Eberhard and of Garg and Mermin [28][29], which the loophole-free tests of 2015 close for nature [30][31][33].

**Discrete gravity and its tests.** Regge calculus [35] and causal sets [36] discretise the geometry of space-time itself; here the lattice is never curved, and the weak-field forms are read from the coefficients (Section 8). In the parametrised post-Newtonian framework [38] the one assumption gives  $\gamma = \beta = 1$  exactly in the model's own potential, with the matter band's factors against the potential the tests read closing with its gap. A lattice breaks Lorentz invariance at order one unless its terms are tuned [39], and the bounds are reviewed in [40]; the paper's answer is no loops and the Link's bound of S.24, with the closing gap for matter (Section 8).

## 3 The model

This section states the model: the lattice, the families, the rule, the paces through which one family acts on another, the one write, the readings and the order of acts in an interval [1, the GameBoard and the families; Rule3; the primitives].

### 3.1 The GameBoard

The GameBoard is a box of  $N_x \times N_y \times N_z$  Nodes. Two Nodes that differ by one step along one axis are joined by a Link, and each Node has six Links, one through each of its six Ports,  $\pm x$ ,  $\pm y$  and  $\pm z$ . Each axis is declared periodic, open or folded. Beyond an open face no Node lies and a level there is read as 0; a folded axis has one layer, and its two Ports return the Node itself. An inner face is a plane of Nodes declared closed except for declared gaps, which is how a wall with slits is made. A face may also be declared receding (ASSUMPTION): the box then grows by a layer of zeros whenever a nonzero level reaches the Node next to the face, so no event ever meets the face, and a run on such a box is a run on the unbounded lattice. The box grows and nothing stretches, so a receding face gives no cosmological redshift [1, the objects; the unbounded board][3, the receding face].

**The group and its action.** The symmetry group of the cube has 48 elements, the 24 rotations and, with the reflections, the signed permutations of the three axes,  $O_h$ , the octahedron's group as it is the cube's (Fig. 1(b)); with the translations it is the space group of the simple cubic lattice. It acts by permuting the Nodes and their Ports and carrying every level pair with its Node; the time reversal exchanges a record's present and its past and carries the step to its inverse (Theorem 1). Rule3 commutes with the space group's action, since its line reads the six Ports symmetrically and names no axis. Under the group the form  $D$  is a scalar, even under the time reversal; the current  $F$  a vector, odd under it; the Wronskian  $W$  a scalar under the cube's 48, odd under the time reversal and under the plane's reflection; and the click a scalar, naming a region and an interval and never a Node or an axis. No boost, no rotation beyond the cube's 24 and no scaling is in the group, so Lorentz's group and the continuum's rotations are no

symmetries of the lattice, and what the paper finds of the moving clock it finds as a property of the clicks (Section 8) (THEOREM for the covariance and the readings' kinds; GAMEBOARD; S.4) [1, the direction; the four acts][3, the one algebra].

The NodeState holds the model's own numbers and nothing else: for every family and every part, two levels and one remainder, and for every part of a field family, the remainder of its one write. A count, a current, a charge or a tension is a reading computed from these numbers when it is needed; no register, table, formula or random draw is kept at a Node [1, the objects][3, nothing at a Node but the law's numbers].

### 3.2 Families and records

A family is declared by two keys: its pair [num,den] and its shape. A family of quanta, such as matter or light, declares its dimension, 1 for one real line or 2 for a plane of two lines, and holds nothing. A family of dimension 3, three real lines with a rotation among them on each Link, is declared by no file and enters only as an explicitly named hypothesis of Section 10.3 (HYPOTHESIS). A field family declares its sources, the form, the form with the three tensions, or the Wronskian, which fix its parts, one real line per source, with its divisor  $E_s$  and, for the massless field, its rest; a field sourced by the form or the tensions holds the content, which slows the clocks of the families that read it, and a field sourced by the Wronskian holds the sign, the charge's field. A family's record is its levels over the GameBoard, at every Node a level now and a level one interval before, with a remainder; a family of dimension 2 lays two level pairs (re,im), two solutions of the same line, and only a plane carries a charge (Section 5.5). A body is a connected set of Nodes on which a family's record stands bound, and a message is a travelling record laid by the world [1, the objects].

### 3.3 Rule3

At every Node, in every interval, every family steps by the first line of Eq. (1). The six arrivals through the Ports, each weighted by its Link's coefficient  $R_{ij}$ , the Node's own level times  $S$ , minus  $w$  times the level before, plus the remainder the Node kept, are summed as one whole number and divided once by the wall  $w$ ; the quotient is the next level, and the remainder,  $0 \leq r' < w$ , stays at the Node and never travels. Nothing is rounded away, and the step back is the same division, which is why every run is exact and reversible to the bit (Theorem 1). The name comes from the three axes, one coefficient per direction serving its two Ports, the wall's factor 6 the three directions twice; for light the next level is the sum of the six arrivals over 3, minus the level before [1, the line].

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

whose rotation at the wave vector  $\mathbf{k}$  is  $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$ . At wave number zero  $\cos \omega_0 = \text{num}/\text{den}$ , and for the massless pair [1,1] along one axis  $\cos \omega = (2 + \cos k)/3$ , so  $\omega = |k|/\sqrt{3}$  as  $k \rightarrow 0$ , and along the cube's diagonal  $\cos \omega = \cos k$  exactly, no dispersion (DERIVED, GAMEBOARD; S.1). Light's speed at long wavelength is therefore  $1/\sqrt{3}$  Links per interval, below the causal bound of one Link per interval, and its group speed falls at shorter wavelengths as  $1 - 0.083 k^2$  [1, the lattice constants]. In nature's names the vacuum's line is the discrete Klein-Gordon equation, exact but for the remainder term: with  $S_6 = 6 + \Delta_{\text{lat}}$ ,  $a_{\text{next}} - 2a_{\text{now}} + a_{\text{before}} = c_K^2 \Delta_{\text{lat}} a_{\text{now}} - m^2 a_{\text{now}} + (r - r')/w$ , with  $c_K^2 = \text{num}/(3 \text{ den})$  and  $m^2 = 2(1 - \text{num}/\text{den})$ , the rest rotation as the mass ( $c_K$  is the equation's coefficient. The band's kinetic scale  $c_m^2$  of Section 8, (e), is  $\omega_0/(3 \tan \omega_0)$  and its largest group speed  $c \cos \omega_0$ , the three one as the gap closes). Its slow limit at small wave number is Schrödinger's equation with the inertia  $m^* = 3 \tan \omega_0$  and, in a content  $U$ , the potential  $\omega_0 f(\omega_0) U$  of Section 7.4 (DERIVED, GAMEBOARD). And  $m^* c_m^2 = \omega_0$  exactly for every pair: the model's  $E = mc^2$  at the band's kinetic scale, read in three click numbers, nature's its small-gap limit, the departure  $\omega_0/\tan \omega_0$ ,  $1 - \omega_0^2/3$  to the order  $\omega_0^2$  (DERIVED, CLICKS; S.18, S.21). At the vacuum's pace Rule3 is the leapfrog scheme of the Klein-Gordon equation, whose dispersion is the known one [41]; what the law adds is one rational [num,den] fixing mass and speed together, the equation exact, its departures predictions with bounds (Section 10.2), the integers with the remainder, and the paces (DERIVED).

### 3.4 The paces: how a field acts on a record

A field acts on a family by changing the family's paces at a Node, the coefficients of Rule3 there:

$$\boxed{\begin{aligned} p_0 &= \Gamma (1 - 1/\Gamma)^c, & p_i &= \frac{p_0 (c_i)^2}{\Gamma}, & p_a(i, j) &= \frac{p_i q_{ij}}{\Gamma}, & q_{ij} &= \Gamma - t_a(i, j), \\ & & c &= \sum_{\text{reads}} \text{weight} \times \text{the read family's time level}, \end{aligned}} \quad (3)$$

Here  $c_i$  is the content the family reads at the Node (the symbol  $c$  names the content in this section and light's speed elsewhere), the sum over its reads of the weight times the read field's level. The clock  $p_0$  composes, each unit of the level slowing it by the same factor  $1 - 1/\Gamma$ , to the unit. The Node's pace  $p_i$  is the clock twice,  $p_0^2/\Gamma$ , so that the ruler goes with the clock;  $t_a(i, j)$  is the Link's tension, one number per Link from its two ends' parts (Section 4.3), read into the Link's factor  $q_{ij}$  the same from both ends. And the Link's pace is the product,  $\Gamma - 2c_i - t_a$  to first order [1, every row reads the content; the clock is the Node's, the tension is the Link's]. A positive weight makes a hollow, an attraction; a negative weight makes a hill. The pace is the clock's slowing; a record at a Node of pace  $p$  rotates and moves as at  $\Gamma$ , with its intervals  $p/\Gamma$  as long. Every family reads the holders of the content its declaration names at the weights it names, gravity's and the binding field's at the weight 1 for every family, the fields among the readers and a holder of the content its own level among it (the law's line, implemented; Section 3.7; the nuclear holders of Section 10.3 by the nucleon families alone), so  $x = (L_b + L_g)/\Gamma$  with  $L_b$  the binding field's level and  $L_g$  gravity's. A family of dimension 2 also reads the holder of the sign plainly, and light, the holder's own record, reads the holders of the content as every family does and not the holder of the sign. And a family sources what it reads at the weight it reads with, so every pair of families that read each other is reciprocal [1, the paces; the reads are the held row's pair's].

**The one assumption of the forms: a Link is clocked twice by its reading Node's own clock.** Nothing in Rule3 fixes how a level enters the paces; beside the line the model holds that a level is a count of units and every act is measured in the Node's own units, its second input beside the line (ASSUMPTION). Two parts of the paces follow from that. The clock composes,  $p_0 = \Gamma(1 - 1/\Gamma)^c$ , each unit slowing the clock as it stands by the same fraction, since the write of one unit is an act measured in the Node's own units, the read included (DERIVED from it). And every act of the interval, Rule3's step, the held rows' write and the sign holder's turn, is Rule3's act measured in the cell's units, so a uniform level  $U = c/\Gamma$  changes only the units, time by  $N = p_0/\Gamma$  and length by  $h = p_0/p_a$  per Link, the write carrying  $p_x p_y p_z / (p_0 \Gamma^2)$  on the form and  $p_x p_y p_z / (p_0^2 \Gamma)$  on the Wronskian, and the turn  $p_0/\Gamma$ , per proper volume and per proper interval (DERIVED, S.34; implemented, Section 3.7). The one sentence not in Rule3, the model's one assumption of the forms, named as the credit is (Section 5): a Link is clocked twice by its reading Node's own clock,  $p_a = p_0^2/\Gamma$ , that is  $hN = 1$ , the ruler going with the clock (ASSUMPTION). The lattice says why twice, a Link having two ends, and not why the reading Node's clock stands at both rather than the two ends' product, which gives  $\beta_K = -1$ ; the sentence is fixed by two clicks, the bending's  $\gamma = 1$  and the perihelion's  $\beta = 1$ , and this is where Einstein's  $\gamma = \beta = 1$  enter the model by hand (ASSUMPTION, FITTED: an input chosen with two known clicks in view; every form of Section 8 is conditional on it and none is a consequence of Rule3 alone; the bending and the perihelion fixed it and are no test of it, while Shapiro's delay, the redshift's factor  $f(\omega_0)$  and the shadow's  $2em$ ,  $e$  Euler's number, remain its predictions). Under it the model's static metric for light is  $ds^2 = -e^{-2U} dt^2 + e^{2U} d\mathbf{x}^2$  to  $O(U/\Gamma)$  and in the ray limit, the exponential metric [42], Einstein's to the first post-Newtonian order; the clock's rate is  $N$  in the lattice's variable  $2\sin(\omega/2)$  exactly and in  $\omega$  to the finite-rotation correction  $\omega_0^2/12$ , 5.9 percent at the computed pair (6.3 exactly) and  $10^{-20}$  at nature's LHAASO bound ( $4 \times 10^{-29}$  at the axis bound), the metric stated in the variable in which it is exact (DERIVED; S.31); the shadow's capture radius  $2em$  its second-order click against Schwarzschild's  $3\sqrt{3}m$  (Section 8), in the lattice's frame, the frame a second residue not derived from Rule3 and read by the binary pulsars'  $\alpha_1$  (Section 10.3). The quadratic  $(\Gamma - c)^2 + c^2$  is the composition's second-order truncation [1, the same clock and the same ruler for every act; the paces compose][1, what is open, item 15].

**The band at a pace.** From the coefficients, the rotation of a record at a Node of clock pace  $p_0$  and Link paces  $p_a$  is (DERIVED, GAMEBOARD; S.1)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a), \quad (4)$$

The mass term scales with the clock's square and the band term with the Link's. Two readings follow. First, at  $k = 0$  every massive band's rest rotation obeys  $1 - \cos \omega_b = (p_0/\Gamma)^2 (1 - \cos \omega_0)$ , the clocks' shifts of Section 7.4. Second, along a Link every wave's speed falls with the Link's pace,  $\Gamma - 2c - t_a$  to first order, so light is delayed and bent by twice the clock's share. At a Node of content  $x = L/\Gamma$  light's band is  $[2 - 4N^4, 2]$  in  $2\cos \omega$  with  $N = (1 - 1/\Gamma)^L$ , and at long wavelength  $\omega = N^2 |k|/\sqrt{3}$ : light's speed is  $N^2 = e^{-2x}$  times its vacuum speed, to the unit, an index  $n = e^{2x}$ ,  $1/(1 - 2x)$  to the first order, with  $n - 1 = 2x$  (S.2). A record whose rotation lies outside the band at a Node is evanescent there and reflected whole; light meets a total mirror where  $p_a < \Gamma \sin(k/2)$  [1, the band at a pace; the rows against nature, (b)].

**The guard.** A pace is positive by the composition and must lie below an edge  $P$ , with  $P^2 = 2 \text{den} \Gamma^2 \text{div} (\text{den} + |\text{num}|)$ , which is  $\Gamma^2$  for light: beyond  $P$  the mode at wave number  $\pi$  on every axis grows without bound, and at or below  $P$  no mode grows exponentially (THEOREM, GAMEBOARD). The guard is the Courant, Friedrichs and Lewy condition of

the explicit scheme Rule3 is at the vacuum's pace [41], checked once at load. Under the composed paces no Link ever closes: the clock  $\Gamma(1 - 1/\Gamma)^c$  is positive at every content and the Node's pace is its square over  $\Gamma$ , so a deep well freezes a clock and closes nothing, and the quadratic form's horizon at  $\Gamma/2$  is a feature of the truncation and not of the law (DERIVED, GAMEBOARD; S.3). A Node whose Link pace rounds to 0, a frozen Node, reads nothing: no event enters it; its level is kept, matter's rotation nearly stopped and light's drifting linearly, and its active neighbours still read it at their own pace, so the cut is one-way; its share is 0 by the instrument's declaration, so it is dark (S.3, S.41). Two thresholds carry the word: the Link's pace at  $(\Gamma/2) \ln 2\Gamma$  levels, the engine's 28,206, and the clock's at  $\Gamma \ln 2\Gamma$ . The guard's bound,  $|2 \cos \omega| \leq 2$  at the band's edges, admits the edge itself, where a linearly growing solution exists, and it is stated for  $\text{num} \geq 0$ ; at that edge the vacuum's checkerboard mode, driven by the remainders, grows as  $t^{3/2}$ , no click reading it, and reaches the loader's amplitude bound, 9,266 levels on the shipped files (GAMEBOARD), a negative numerator's lowest rotation lying at  $k = 0$  (S.3). At a negative content light's zone-corner mode grows, so light refuses a content below 0, and the vacuum's content is 0 (Section 6.5) [1, the guard; the frozen Node has no share].

### 3.5 The write, the readings and the interval

**The one write.** Every write into a field's level is one act, the second line of Eq. (1): at a Node in each interval a held part gains  $(\sum_f w_f q_f + r) \text{div } E$ , one division per part with one remainder  $r$  carried at that Node. Here  $q_f$  is the sourcing family's own quantity at the Node: its form  $D_f$  for a holder of the content, its Wronskian  $W_f$  for the holder of the sign, or its tension  $T_{aa,f}$  for an axis part.  $w_f$  is the weight with which that family reads the written field, and  $E$  is the part's wall,  $E_s T$  for the time part and  $E_s W_c$  for an axis part. The sources are summed in the numerator and never divided apart, so no remainder stands per sourcing family [1, the primitives][3, the one algebra].

**The readings.** A reading is a bilinear form of the levels with declared coefficients; it writes nothing on the GameBoard. The law reads three, the third line of Eq. (1). The form  $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$ , the present against the product of the future and the past, is  $B^2 \sin^2 \omega_b$  at every interval for a standing record of peak  $B$ ; its quotient over  $T$ , the well, sources the fields. The Wronskian  $W_i$  is the charge (Section 5.5). The current  $F_{ij} = \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$  into the Node  $i$  from its neighbour  $j$  is antisymmetric and, with the Link's factor  $q_{ij}^2/\Gamma^2$ , is the Link term of the conserved form (Theorem 3), the plain current where no tension stands; a wave travelling from  $i$  to  $j$  has  $F_{ij} < 0$ . A static record has  $\text{now} = \text{next} = \text{before}$ , so its form and its current vanish: a static field carries no quanta and no detector sees it [1, the booking; the conserved form].

**The interval.** Every read is of the values at the interval's start; Rule3's line reads one Link per interval, and nothing in zero time; every reading of the interval, the step, the tensions and the hold's write, is read from the record as it stands at the interval's start, so a Node's NodeState after one interval depends on the Node and its six neighbours alone, the interval's dependency radius one Link (the law's line, GAMEBOARD) [1, the interval's dependency radius is one Link]. The interval has four acts in order: (i) the read, each family's paces from the field levels, the reading field's own among them, and each Link's factor from its tension; (ii) Rule3 on every record; (iii) the readings, each record's currents, tension, form and Wronskian at the levels the step started from, and each detector's report; (iv) the write of each field from its summed sources. [1, the interval][2, the interval].

**What is conserved, and where.** On the GameBoard the conservations are per family: the weighted form of a family while its reads are static (Theorem 3), the Wronskian of a plane family under a static level, exactly in the line under the implementation's present convention of the turn (Section 5.5), the quanta through the Ports, the wave number of a free record and reversibility bit for bit. Energy is conserved within a family by theorem, between a charged body and its light by the law's line (Section 10.1), and between a body and the holders of the content it is replaced by the mass defect (Section 10.3). The GameBoard conserves no total: the sum of the families' forms is not conserved and not assumed, the share moving with the paces with no current to compensate (Eq. (13)). The click layer has books of its own: a click credits one whole quantum at the detector's own whole, drawn by the inflow of the record's share through its front over its window; a frequency of nature is read as the inflow's beat within the window or as the absorber's resonance; the GameBoard's quantities, the form, the content, the level, the rotation, are read by no click except as the share's inflow; and the credit's bookkeeping is that the shares sum to what the screen saw, one whole per click, the whole the detector's own transition (Sections 5.3 and 7.4). In the clicks there are conservations other than the board's, and neither layer keeps a total energy (THEOREM for the board's, ASSUMPTION for the credit's) [1, the conserved form].

### 3.6 The postulates

The model rests on the law's seven postulates [1, the postulates], the detector's write added by decision with its gate pending [3, the two hands close the line of the click's write]: (1) the world is Nodes and Events, an Event a level

stepped by Rule3 with its remainder, a click a detector's report and, by that decision, its one write at the Node the quantum reached it from, and nothing else happens; (2) a Node holds bounded local information, read from itself and its six neighbours, and nothing of the past is kept beyond the one level before; (3) consistency is local and causal, one Link per interval for every reading of the interval (Section 4.3), with one exception, the credit of a click by the shares over every declared detector, the one non-local element, not derived; (4) the causal speed is one Link per interval; (5) every physical calculation is on bounded integers, with no floating-point number, no root and no random draw in the GameBoard's step, the detector's draw the instrument's, with its declared seed; (6) a measurement is a detector's click, the net current into a declared region through its front boundary over one interval, accumulated over the window, never a Node; (7) the GameBoard, where no one measures, is deterministic and reversible to the bit, a detector's write the instrument's act, forward only, outside the reversal, and the detectors' clicks are the only thing claimed to represent nature. The line and its clock are fixed: every mend of the model is a family, a holder, a declaration or a read, never a change of Rule3 (ASSUMPTION) [3, nothing that is not Rule3 remains].

### 3.7 The implementation

The implementation, Universe24 [43][2], is the law's implementation and nothing else: a GameBoard of Nodes, every family's NodeState at every Node, and one interval of acts in the law's order, each a call of Rule3 on whole-board arrays of integers at a declared width, every neighbour read through a Port. It holds no number of physics, no formula, no family's name and no flag, every physical value coming from the run's files; a run is headless, writes the click lines, the measurement, and runs for results only after the check of Section 4.1 says MATCH [2, the engine; the NodeState; the output; the gates]. It holds the law's lines of Section 3.4, and the checked worlds' outputs under them are bit-identical to the quadratic forms' runs [2, the NodeState; the interval]. The checks' runs: the two slits, the Huygens prediction 273, written before the run, the run  $N = 278$ , and the law's real line 277.8 (278.0 in the physical labels), computed after the run by the script `two.slits.real.line.py` beside the paper; Bell,  $S = 478/169$  with the marginals  $1/2$ ; GHZ, the eight shares, every pairwise  $E 0$  and  $M = -4$ ; and the back-in-time check MATCH on every world (COMPUTED, runs of the implementation; CLICKS) [1, the rows against nature, (g) and (h); the GHZ gate]. The paper reports the checks' runs and derives the rest: a run of a declared world computes what the algebra of that world states, the implementation held to the law's lines by its tests and the gate. The implementation is public, with the documents, the files and the scripts, in the Git repository <https://github.com/Closer24/Universe24>, archived at Zenodo [43]: it installs in a clone with `python -m pip install -e .`; the two slits' world runs with `PYTHONPATH=src python tools/run_inputs.py --out runs/first examples/events/two.slits/two.slits.json` and is read with `tools/click_counts.py`, Bell's and the GHZ's worlds with `tools/bell_gate.py`, and the back-in-time gate with `tools/back_in_time.py`, each as the engine's document states it [2, how to run a world]; the paper's own scripts stand in `paper/general_formula/`.

**The rules we took.** What this paper computes is one chosen set of rules: Rule3 with the one write and the three readings (Eq. (1)), the postulates of Section 3.6, the primitives the files declare by name (the families with their pairs and dimensions, the holders with their sources and divisors, the detectors with their four declarations, the faces) and the integers of `rule.json`,  $\Gamma$ ,  $T$ , the width and the Link unit (Section 6.6). The engine fixes Rule3's form, the recurrence, and nothing else of the set: another universe file, other families and primitives, other integers can be taken and other things built with it [2, the main loop and Rule3; how to add a family]. Everything the paper derives is derived from this set and claims nothing beyond it; where a derived form coincides with a known one, the text says derived and names the form it meets, and never presents the known form as the law's own.

## 4 Exact properties

This section states what holds exactly on every lattice, each proved in S.48: the reversibility of the rule, the kind of function its acts can form, the conserved form with its current and its momentum flux, and the count as a reading of the record (S.5, S.6).

### 4.1 Reversibility

Let  $\sigma$  be the direction,  $+1$  forward and  $-1$  backward; let `div` and `mod` be the floor division and its remainder in  $[0, w)$ ; and let  $\rho$  be the remainder carried in. With

$$u = \sigma \left( \sum_j R_{ij} \text{arr}_j + Sx - wy \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (5)$$

the forward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$  gives  $(z, \rho') = (a_{\text{next}}, r')$ , and the backward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$  gives  $(z, \rho') = (a_{\text{before}}, r)$ .

**Theorem 1** (the direction). *The backward call returns  $(a_{\text{before}}, r)$  bit for bit; the reversibility of every step needs no second function.*

(Proved in S.48.) Theorem 1 holds for the GameBoard's step at every interval; a detector's write and a body's window, click or not (Section 5.4), are the instruments' acts outside the reversal, as the lay is; the back-in-time gate is exact between clicks and runs across a click from the output's click line (THEOREM for the step; the write ASSUMPTION). The whole run therefore goes back:  $N$  intervals forward and  $N$  back return every array of every family bit for bit, provided every number an act read at an interval's start is one the state after the interval still holds (THEOREM where the writes change no level, by construction from Theorem 1). The implementation checks this as a test, running a world forward and back and comparing every array at every interval; on the worlds of Section 9 it reports MATCH over their whole runs, worlds whose write stays under one level (Section 6.6) and whose detectors write nothing; the interval's reversal with a level-changing write is not yet gated (COMPUTED, GAMEBOARD) [1, the inverse; the objects][2, how to run a world][3, the Node knows no family]. The condition rules out one read: a coefficient read from a Node's own state across a step cannot be recovered at the step back, so a record's rotation sense is never read at a Node and the holder of the sign is read by the record's shape, a plane or a line (Section 5.5) [1, the interval][3, the board goes back in time, as a gate; there is no sense].

## 4.2 What the acts can form

Rule3's line is one act, and declared coefficients make it four: (a) the read, the line with its six arrivals; (b) the one-Node step, the line with the arrivals declared 0, which rotates a level by Chebyshev's recurrence; (c) the division, the line with  $R = S = 0$ , Euclid's division with the remainder kept; and (d) the load, a declared integer in the numerator, which is how a count enters a level [1, the four acts].

**Theorem 2** (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

(Proved in S.48.) So every product of two levels is a reading and never a write, and the three readings of Section 3.5 are the only products the law takes. A line enters the law only if it passes three tests: it is generic (one primitive with declared integers, no family's name), vector (an act on the whole state, with no root and no floating-point number) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers). A line that needs more is a hypothesis under its own name. No act of the interval takes a square root; the integer square root appears only when a world is laid, as the fixed point of the division act  $x \leftarrow (x + n \text{ div } x) \text{ div } 2$  [1, the three tests of every rule][3, the six verbs; the root leaves everywhere].

## 4.3 The conserved form and the momentum flux

The read coefficient  $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$  carries the Node's factor and the Link's; the Node's factor is not symmetric where the paces differ and the Link's is, so the conserved form carries the weight  $1/p_i^2$  on its Node terms and the Link's factor  $q_{ij}^2 / \Gamma^2$  on its Link terms:

$$E = \sum_i \frac{w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i}{p_i^2} - 2 \text{ num } \sum_i \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} \text{now}_i \text{before}_j. \quad (6)$$

**Theorem 3** (the conserved form). *At constant paces for which a positive Node weight  $d_i$  with  $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$  on every Link exists,  $E$  with the weights  $d_i$  is exact under the line without the rounding. Under the law's read coefficient, the Node's pace squared times the Link's factor squared,  $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$  with  $q_{ij} = q_{ji}$  and  $p_i = p_0(c_i)^2 / \Gamma$  the composed Node factor, the weight  $d_i = 1/p_i^2$  cancels the Node's factor on both sides of every Link, so  $E$  is exact for any static field and static tensions. Its Link term carries the Link's factor, so the current of Rule3's own conservation law through the Link  $ij$  is  $\text{num}(q_{ij}^2 / \Gamma^2) (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ , the plain current  $F_{ij}$  where no tension stands,  $q_{ij} = \Gamma$ ; the form with the plain current holds there alone, and with a tension it is not conserved (two witnesses, S.32).*

(Proved in S.48.) With the rounding each Node's remainder adds the term  $(r - r')(\text{next} - \text{before})/p_i^2$  to the walk of  $E$  (DERIVED; S.5).  $E$  is a diagnostic. Where the paces move, the total form  $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$  changes exactly by  $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}} \rangle$ , which is 0 at fixed uniform paces and negative where a well deepens under a standing record (THEOREM, by substitution; GAMEBOARD) [1, the conserved form].

**The momentum flux is the wave's own stress.** The momentum density along the axis  $a$  at the Node  $i$  is the current per axis,  $P_a(i) = (F_{i,i-a} - F_{i,i+a})/\text{num}$ , and the line without its division gives exactly, at constant paces, the identity below; the integer step adds the remainder term  $\delta_i \Delta_a(\text{now})_i - \text{now}_i \Delta_a(\delta)_i$  with  $\delta = r - r'$  and  $\Delta_a$  the centred difference, which the implementation's test includes (THEOREM for the line, GAMEBOARD),

$$w [P_a(t+1) - P_a(t)] = \sum_b R_b [G_{ab}(i) - G_{ab}(i-b)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

with  $G_{ab}(i)$  the stress reading on the  $b$ -Link ( $i, i+b$ ), the levels now alone. The proof is in S.48. The tension on the axis  $a$  at a Node, the part a field reads into the Link's pace and the source of a field's axis part, is the momentum flux averaged over the Node's two  $a$ -Links,

$$T_{aa}(i) = -\text{num} (G_{aa}(i) + G_{aa}(i-a)) \text{div } 2, \quad (8)$$

$$G_{aa}(i) = \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}),$$

and each Node sources its own part,  $-\text{num } h_a(i)$  with  $h_a(i) = \text{now}_{i-a} \text{now}_{i+a} - \text{now}_i^2$ , read from its two neighbours along the axis at the interval's start; the Link's tension is its two ends' parts, one number per Link read the same from both ends by the pace at the next interval. A plane wave of amplitude  $b$  at the wave number  $k$  has  $G_{ab} = -2b^2 \sin k_a \sin k_b$ , so the flux along the wave,  $T_{aa} = -\text{num } G_{aa} \geq 0$ , is positive, as a wave's pressure is (S.7); a packet of count  $c$  and velocity  $\mathbf{v}$  has  $2cv_a v_b$ , the two Links' currents summed, the stress of dust (S.7 for the plane wave). With the sign as written a wave's stress deepens the content around it, so light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification][3, the tensor part is the record's own stress].

#### 4.4 The count is the record's share

A family's count of quanta is a reading of its record and no line of its own. At every Node the record's share is

$$e_i = \left[ w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i - \text{now}_i \sum_{j \sim i} R_{ij} \text{before}_j \right] \text{div } (2p_i^2), \quad (9)$$

the conserved form's term at the Node with its weights (Theorem 3), the Node's  $1/p_i^2$  and each Link's  $q_{ij}^2/\Gamma^2$  inside  $R_{ij}$ , read in quanta over  $W_c$  as the nearest whole number; at the vacuum's paces it is  $3 \text{den} (\text{now}_i^2 + \text{before}_i^2) - \text{num now}_i S_6(\text{before})_i$  exactly,  $S_6$  the six-neighbour sum, and read with the vacuum's coefficients instead of the Node's own paces the same expression is the vacuum's share, the unit of the tables of Section 7. The implementation reads it so. Over one step of Rule3 the share's rational value, before its floor, changes by exactly the six currents into the Node, each weighted by its Link's factor, plus Rule3's own rounding (THEOREM for the rational identity, GAMEBOARD; S.5); the integer share of Eq. (9) adds the reader's own floor, a residual strictly between  $-1$  and  $1$  per Node and level pair, distinct from Rule3's remainder:

$$e_i(\text{next}, \text{now}) - e_i(\text{now}, \text{before}) = \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} F_{ij} - \frac{(\text{next}_i - \text{before}_i)(r'_i - r_i)}{2p_i^2}, \quad (10)$$

with  $F_{ij}$  at the levels the step started from and  $r, r'$  Rule3's remainder before and after the step. So over any region the total share changes only by the weighted currents through the region's boundary, the remainder terms and the readers' floors (THEOREM, GAMEBOARD) [1, the count is the record's share; the share's change is the currents]. A formulation moving whole quanta by a division of the current would put numbers at every Node that no act of Rule3 reads back; the law does not carry it (ASSUMPTION) [3, the count's line and the sense's line leave the law]. Where the paces move the share drifts with no current crossing any Link, Eq. (13), a diagnostic [1, the paces' own change moves the share with no current].

**The signed integers.** The levels are signed integers, the currents are signed by their direction and the Wronskian by its sense, and a click is one whole quantum, never negative. The total share of a closed or periodic GameBoard is non-negative for  $|\text{num}| \leq \text{den}$ , with equality for a static uniform level of light and, on an even periodic board, for its checkerboard (THEOREM, GAMEBOARD; S.6). One Node's share can be negative at a lay's edge, and a region's by at most half a level-square per boundary Port. Over the whole passage of one forward packet the inflow accumulated at a region's front equals the region's final share plus what left through its back, each bounded below by S.6 (Eq. (10)

summed over the window), so it is non-negative up to the rounding and the credit by the shares is well defined; where a window is cut inside a passage the instrument credits  $\max(s_R, 0)$  (DERIVED, CLICKS) [3, the law's integers are signed].

## 5 Measurement: the click as a meeting

The click is the meeting of the future with the past at a declared detector, and no act of the GameBoard. A click is a fact: one whole quantum passed from a giver to a taker, written once and never undone; it is always the meeting of two quanta, so every click has its giver and its taker, the record and the body's transition, and happens never by itself (S.49 (16)). This section states what is on the lattice, the meeting, the detector, what a click leaves behind, the charge and light, and what locality costs [1, the click; the reading].

### 5.1 Events, and no waves

On the GameBoard there are events and amplitudes. An event is a level split by Rule3 to the six neighbours; an amplitude is the level Rule3 produces at a Node, the sum of the split events there; and the record is the amplitudes over the GameBoard. In our world there are clicks, the detectors' reports, and nothing else. In the two slits the emitter's events pass both gaps, every path of the future at once; the fringes are where split events from the two gaps meet at the screen with the same sign. The past is made of the same split events: read backward from the realised click by the same line, Rule3 at  $-1$ , they form the past's tree, which also passes both gaps back to the emitter. What stays on the GameBoard, the amplitudes no click realised, is not in our world: after a click they are a future that cannot be realised, an empty wave, a GameBoard diagnostic (Section 5.4) (ASSUMPTION) [1, no waves on the event board][3, no waves on the event board].

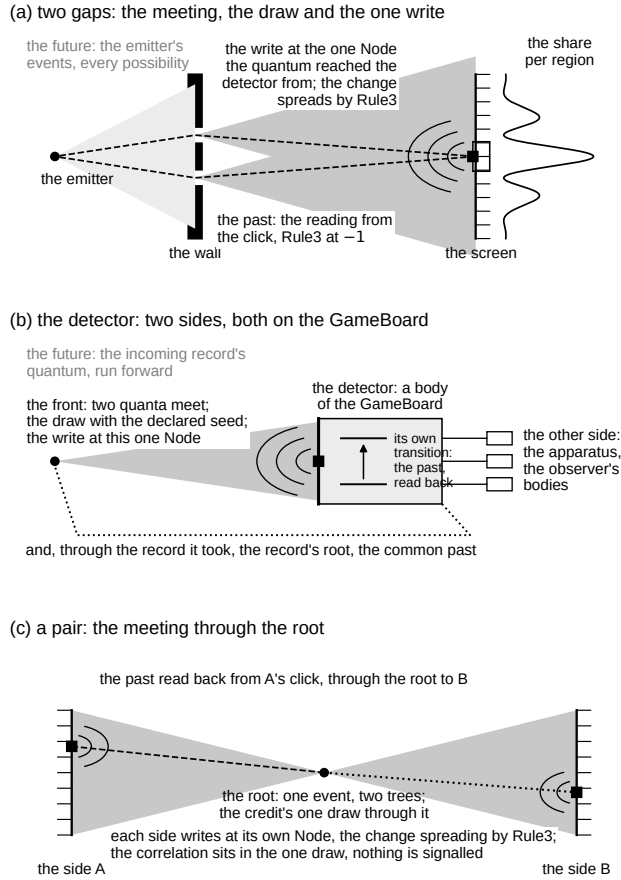
### 5.2 The click is the meeting of the future with the past

**Assumption 1** (the click is a meeting, the law and not a hypothesis). A record runs on the GameBoard in both directions by the one line: forward from the emitter, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at  $-1$ . A click is the meeting of two quanta at a declared detector: the incoming record's quantum run forward, the future, and the detector's own transition, the click's unit, the past read back from it; its share is their product at the region, at resonance (Section 7.4), the form  $D = \text{now}^2 - \text{next} \times \text{before}$ , the now against the future times the past; which meeting is realised is the detector's draw with its declared seed, and the GameBoard draws nothing; the one non-local place is the credit's one draw over the detectors that read one record, through the root. The detector only asks whether the future and the past met at it.

Two detectors stand under the one word: the implementation's declared region, a bolometer's front with no record of its own, on which every check reported here ran (Section 5.3); and nature's detector, a body of the GameBoard whose front is such a region and which has its own transition (Section 5.4).

**The click, algebraically.** Let  $X$  be the state space: at every Node, for every part of every family, one integer level pair and the write remainders, and nothing else. The step  $S_t$  is Rule3 over the interval with the paces read from the fields at its start, a bijection of  $X$  (Theorem 1); its forward orbit is the future and its inverse orbit the past. An instrument  $R$  is a declared region with its front Ports and its coefficients, which define  $Q_R$ , the inflow through its front over the window (S.50). A click is the pair  $(R, t)$ , the credit of one whole quantum to  $R$  at the interval  $t$ , drawn with the probability  $s_R / \sum_{R'} s_{R'}$ ,  $s_R = Q_R(x_t) / W_c$ ; the decision is the one map in the model that is not a bijection, a scalar under the group, so no click names a Node to us; its write is one act on  $X$ , at the one Node of the region from which the quantum reached the detector (Section 5.4): the levels that arrived there are taken, the record's count falls by one and the taker's rises by one, the quantum conserved between them, and the change spreads by Rule3 as an event (Section 5.4; ASSUMPTION, the instrument's declaration). A frozen Node stands in no instrument's front and is never clicked (ASSUMPTION for the definition, DERIVED for the precisions; CLICKS) [1, the click's algebraic definition].

**One form for every click.** At the credit the parts of a record are read with the detector's coefficients, each part paired with the same part through the common origin, summed and squared; the share is that density, and the draw is by the shares. The two slits are the case with nothing to pair: light is one real line, and the share is the record's own inflow. Bell's check is the case through the common origin: a pair born as one event is two trees with one root, and the backward reading from the first click passes the root and continues forward on the partner's tree (Section 9.3). In the implementation the joint share is computed from the forward records alone, by the linearity of the rule, the check of Section 4.1 showing that the backward reading exists to the bit (the credit ASSUMPTION; the equality of the forward and backward descriptions DERIVED for the line at fixed paces; CLICKS).



**Fig. 2** The click as the paper states it, schematically (ASSUMPTION; drawn from the definitions, no run). (a) The two slits: the emitter's events spread forward through both gaps, every possibility at once (the grey fans, the future); from a realised click at the screen the same line is read backward, Rule3 at  $-1$ , through both gaps (the dashed lines, the past); the share along the screen, the product of the two directions, alternates; the detector writes once, at the one Node from which the quantum reached it (the filled square), and the change spreads from there by Rule3 (the arcs). (b) The detector is a body of the GameBoard with two sides: at the front the incoming quantum meets its own transition, the detector draws with its declared seed and writes at the one Node; its other side is bound to the apparatus, the observer's bodies and, through the record it took, to that record's root. (c) A pair, two trees with one root: the backward reading from A's click passes the root and continues on B's tree (the dotted line); the credit's one draw through the root carries the correlation, each side writes at its own Node, and nothing is signalled.

### 5.3 The detector is a declared instrument

**Assumption 2** (the detector's declaration). A detector is one connected region of Nodes declared in the file, never one Node, at least half the wavelength of the family it reports across the beam (4 Links at  $\lambda = 8$ ); a narrower region, a single Node or a region in pieces is refused at load. It is opaque, backed by a receding face or by the wall's own face, so that what enters it leaves the declared board and never returns. Its click is its report, per interval and family: the net current into the region through its front boundary Ports, the Ports leading in from the declared board, summed in integers, with the region's name and the family and never a Node; one wall  $W_c = 3 \text{ den } T$  of the inflow over a passage is one unit of count, the bolometer's whole ( $W_d = T$ ; a counter's whole is  $T \sin \omega$ , S.49 (14)), so a screen saw  $N$  units where its regions' inflows sum to  $N W_c$ . The credit: one quantum to one detector, drawn by the shares of the regions' inflows, read outside the GameBoard over every declared detector of the world, one credit per quantum and never two; the detector draws with its declared seed, the GameBoard draws nothing and the run is deterministic per seed. After a realised click the detector writes once, at the one Node from which the quantum reached it (Section 5.4). The credit is the one non-local place of the description and no line of the law.

What produces a click is deterministic, the meeting; which of the possibilities becomes the click no line of the law gives, and the instrument draws. Exclusivity, one click per quantum, and Born's rule, the credit's chance equal to the intensity's share [44], are declarations of the instrument and not findings of the model [1, what produces a click is not

known]. The uncertainty relation is the detector's declarations, a region of at least half a wavelength, a window, a floor and a whole, with one draw per quantum (ASSUMPTION) [3, the uncertainty relation is not a rule of its own]; what each declaration costs is derived in S.8 and S.50, a pattern in the histogram and never in one click (DERIVED, CLICKS) [2, the output].

## 5.4 The click decides in the credit and writes at one Node

**Assumption 3** (the click's decision and its write). The GameBoard is deterministic and local: Rule3 with no draw, Theorem 1's bijection at every interval. The click is the meeting of two quanta at the detector, the incoming record's quantum and the detector's own transition; the detector, an instrument declared in the world file, draws with its declared seed whether the meeting is realised, and the randomness is the detector's, the board has none. Forward, to the future, one write: the exchange at the one Node from which the quantum reached the detector, which the detector knows and we do not (the uncertainty principle: for us the click is a region and a window), the one Node drawn in the same one draw by the window's inflows per Node, the region's share their sum, the Node a GameBoard diagnostic beside the click line, as the implementation's round holds it; the detector's record takes its transition, the levels that arrived at that Node are taken, whatever they are, as the receding face takes a share, the record's count falls by one and the taker's rises by one, the quantum conserved between them, and the change spreads by Rule3 as an event, one Link per interval, with no global act; the other paths stay on the board as an empty wave (Section 5.1). Backward, to the past, a reading and never a write: the credit's reading through the root, which chooses which tree is the past and gives Bell at a distance (Section 9.3), the one non-local place; it cannot be a write, the board being deterministic forward.

A single quantum's count stands at 0 after its click, the count conserved and read by the credit through the root, the detectors' shared other side on the GameBoard, the Nodes' levels carrying the shares and never the count: antibunching and one quantum one click between far detectors [45], the credit's one draw over all the detectors that read one record (ASSUMPTION for the one draw, DERIVED from it for the rest; CLICKS; the numbers of Sections 9.2 and 9.4 unchanged, one click per record). No signal passes: nothing at one detector changes the amplitudes at another except by Rule3 from the Node of the exchange. A detector is a body of the GameBoard and stands nowhere outside it: it has two sides, the front where the incoming quantum meets its own transition and the other side where its own record connects to the rest of the board, the apparatus and the observer's bodies [1, the observer is a cluster] and, through the record it took, that record's root, the emitter's event, so that two detectors reading one record are connected through the root, their common past; it can be entangled on both sides. Bound bodies click by quanta from outside meeting their own transition, never by clicks from outside, the meeting the resonance of Section 7.4 (DERIVED, CLICKS); a body reading its own parts draws over them at each window and is re-laid in the drawn part at its count, a click or not, while a detector reading a light record writes nothing on a null outcome (S.59); what clicks: light with a body's transition (the taking and the giving), a quantum of matter with a body's transition (the capture and its reverse), light with light by summing in one row and never by a click, and the holders of the content, gravity and the binding row, never click (S.49; the conversion of a quantum between families the hypothesis below); a body in the dark is only the future and still gravitates; emission is the same meeting with the roles reversed, two local writes and one draw of the credit binding them. The Node stays generic, knowing no family and no click; the write is the instrument's act (ASSUMPTION, the instrument's declaration [1, the click writes on the GameBoard]; the decision's form DERIVED, the one form of Section 5.2). The GameBoard holds only the future, every possibility, and has no arrow: Rule3 is a bijection at every interval (Theorem 1). The clicks have the arrow: each is forward only, the one act outside the reversal; at each click what was future becomes past, the present being the meeting; the chain of clicks is the past advancing into the future, written once and never undone. The direction of time is the clicks' and not the board's (THEOREM for the step; the clicks' arrow the instrument's declaration). The engine round of the click's write is not yet gated, and the checks reported in this paper are those of detectors that report and write nothing (Section 3.7) [3, the two hands close the line of the click's write; the detector only writes, and it writes only to the one Node from which it received the click; on the GameBoard there is no draw; the one who makes this draw is the detector; there is no family of clicks at the Node][1, the click is the meeting of the future with the past; clicks go only forward; the backward run is exact].

**In plain words (Fig. 2).** A click happens when the past meets the future, and is created only at their meeting; this is a line of the model and not a hypothesis [3, the click is the meeting of the future with the past]. The board holds every path that can still happen: the future. A detector is a body on the board. When the quantum coming in meets the detector's own transition there is a click, the present, and the past is what is read back from the click. The detector decides by its own draw and writes once, at the one Node from which the quantum reached it; from there the board carries the change on its own, one Link per interval. The quantum's count is conserved, so it is taken once and no other detector takes it. The other paths are a future that will not happen. The clicks have the direction of

time; the board has none. A bound body, an atom or an ion, clicks in the same way when a quantum from outside meets its own transition, and that is how it absorbs, emits and jumps (ASSUMPTION for the draw and the write, the rest DERIVED; CLICKS).

**A hypothesis under its own name: only a click moves a quantum between families.** On the GameBoard a record's levels change by Rule3 in its family's plane and by the holders' reads alone, and the read moves no form; form passes between a family and a holder's record only through the write at a body, the giving and the taking; a count passes between families only at a click, at a taking the record's count down by one and the body's part up by one, at a giving the body's part down by one and one record of light with count 1 laid; between clicks every count is constant. In the model owner's reading: the GameBoard computes the possibilities of the future locally and carries shares; the clicks are the present, where under the hypothesis the world's real computation is done, each between the past, the body's own transition read back to the record's root, and the future, the incoming record's paths; what passes between families on the board is form, a possibility, and what passes at a click is a quantum, a fact. What tests it: a body laid in its excited part alone, with no beat, gives no light by the write alone and one quantum at an exponential time under the hypothesis, a single atom's lifetime; its gate worlds are named in the open list (HYPOTHESIS [1, only a click moves a quantum between families]; no number of this paper rests on it; Section 10.3).

## 5.5 The sign is the rotation sense

A family of quanta laid as a plane carries its record as two level pairs, (re, im), stepped by one call of Rule3 with one set of coefficients, each with its own remainder; a field's parts stay real. The charge density at a Node is the reading

$$\boxed{W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}}, \quad (11)$$

the Wronskian of the two pairs, which the line without its division conserves exactly with the conserved form's weights, under Theorem 3's condition on the paces (THEOREM, since **DM** is then symmetric; GAMEBOARD; S.9). The integer step adds to each part the remainder term  $\epsilon = (r - r')/w$ ,  $|\epsilon| < 1$ , so the integer  $W$  changes at a Node by  $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$  per interval, not zero in general, one witness computed in S.9 (COMPUTED, GAMEBOARD); an exactly conserved integer charge is open (Section 10.3). Its local change is the Link current,  $(2 \text{ num}/w) (q_{ij}^2/\Gamma^2) (\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$  per Port for the weighted  $W_i/p_i^2$ ,  $\text{num}/(3 \text{ den})$  times the bracket for the integer  $W_i$  at the vacuum's paces (S.9), and its total over the GameBoard is invariant under any change of the paces that keeps them uniform over the GameBoard, the weighted total  $\sum_i W_i/p_i^2$  exact at any paces, since the bracket  $\mathbf{im}_{\text{now}}^\top \mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{now}}^\top \mathbf{M} \mathbf{im}_{\text{now}}$  vanishes for every symmetric  $\mathbf{M}$  (THEOREM, GAMEBOARD). A real record ( $\text{im} = 0$ ) has  $W = 0$ , so light, the real record of the holder of the sign, is neutral; a rotating record has  $W$  of either sign by its sense [1, the sign is the rotation sense; the sign's count is the record's own].

Who reads the holder of the sign is decided by the family's dimension and by no name (ASSUMPTION, GAMEBOARD): a family of dimension 2 has a Wronskian and reads it, a family of dimension 1 has  $W = 0$  and does not. The holder of the sign does not read its own level, so light does not read it and a static charge does not bend light. A reader's own sense never enters its coefficients, since a coefficient read from a Node's own state would break the exact inverse (Section 4.1). So the holder of the sign acts on its readers by the act its row declares: **rotation**, the holder's level over  $\Gamma$  as the angle  $\theta$  turning the two-part record in its own plane,  $e^{-i\theta} z_{\text{next}} + e^{i\theta} z_{\text{before}} = (S z_{\text{now}} + \sum_a R_a (z_{+a} + z_{-a}))/w$ , three exact integer shears inverted bit for bit with  $\theta$  read at now alone, under which like and unlike senses are told apart by the record's own sense. Which sense moves apart is one bit the model does not fix, declared by nature's click that like charges repel, the shears' exact angle  $2 \arctan(L/2\Gamma)$ ,  $L/\Gamma$  at small angles, and the magnetic force follows as the Link's phase, Peierls' coupling (Section 8) (ASSUMPTION for the declared bit, GAMEBOARD) [3, there is no sense; every family has a dimension; the rotation round, built][1, the sign holder rotates the two-part record; the rows against nature, (f)].

## 5.6 Light is born by the write

Light is the holder of the sign's own record: one record whose time part is a real level pair, stepped by Rule3 at the paces of its reads, so that light is bent by the content, and written by the one write from the Wronskian of the records that read it. A static rotating body writes a static level, Coulomb's field, and radiates nothing. A breathing or accelerating body writes a varying source at its Nodes, the field carries it away at one Link per interval, and those spreading events are the light; a uniformly moving body writes no travelling light in the long-wavelength band (S.42 (b)) (DERIVED, CLICKS). A detector reads light's current through its boundary in quanta of  $W_c$ , and the travelling events of the other fields, which in gravity's case nature calls a gravitational wave, enter a detector the same way. A body's quanta that leave it stay its family, and no quantum changes family at a crossing. A body reads the holder of the sign at its own Nodes, so a breathing body reads the light it writes and its pace answers, the radiation reaction, in the same loop [1, light is born by the write][3, light is born by the write].

**Table 2** The formulas of clicks, each with its status. Every row is a formula of what a detector reports, CLICKS.  $U$  is the clock's share and  $U_K$  the Newtonian potential the orbits measure (Section 8).

| The formula                        | Status                                                                     | What the model gives                                                                                                                                                                                                                                                           | Section  |
|------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Born's rule, the count at a region | declaration; $N_R$ and the scatter derived under it                        | $N_R = Q_R/W_c$ , the share of the inflow through the region's front over the window; the pattern is in the histogram, $Np_i \pm \sqrt{Np_i(1-p_i)}$                                                                                                                           | 5.2, 5.3 |
| The dead time                      | declaration                                                                | not derivable: the shares credit a region any whole number of quanta in one interval, and $W_c$ over the inflow's rate is the mean spacing of the clicks and not a least one                                                                                                   | 5.3      |
| The two-slit row                   | derived; computed                                                          | the near-field minima at the rows 15 and 33; the screen's inflow peaks at the interval 68.0 against the fourfold refined lattice's 61.8                                                                                                                                        | 9.2      |
| The arrival and its spread         | derived; computed                                                          | $t = L/v_g$ with $v_g = (\text{num}/(3 \text{ den})) \sin k / \sin \omega$ ; the click's duration $\sigma_t = \sqrt{\sigma_x^2 + (\omega'' L \sigma_k / v_g)^2} / v_g$ from the band's curvature, 4.13 intervals at $k = \pi/4$ over $L = 34$ Links; the centre's arrival 62.2 | 9.2      |
| Bell's pairs                       | the credit declared (Born's rule for a pair); $E$ and $S$ derived under it | $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ and $S = \sqrt{2}(1 + \rho)$ at the ideal angles, 478/169 at the integer settings for equal parts                                                                                                                                 | 9.4      |
| GHZ, three quanta                  | derived under the declared credit (S.40)                                   | $E_3 = \cos 2(a + b + c)$ , every pairwise $E$ 0, Mermin's $M = -4$ where every local credit gives $ M  \leq 2$                                                                                                                                                                | 9.4      |
| The clock's click rate             | derived                                                                    | $1 - f(\omega_0)U$ , $f = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ , $1 - U$ at a small gap; the beat to $(1 - N^2)\omega^2/24$ ; a cavity of bodies the same, declared faces $N^2$                                                                                      | 7.4, 8   |
| The bending's clicks               | derived under the one assumption; in $U_K$                                 | $4U_b$ ; $(1 + \text{den}/\text{num}) 2GM_K/(c^2 b)$ against Kepler's potential                                                                                                                                                                                                | 8        |
| The absorption line                | derived (S.16, S.43); the one-mode line under the pace read alone          | parametric at $2\omega_b$ , the growth $\epsilon/(4 \sin \omega_b)$ ; the transfer line at $\omega_j - \omega_i$ , Rabi's rate, under both reads                                                                                                                               | 7.4      |
| Emission                           | derived                                                                    | a steady rotation writes no light, nor a uniform motion in the long-wavelength band (S.42 (b)); a breathing body writes at its beat, at the first order in the breathing's depth                                                                                               | 7.4      |
| The which-way sum                  | derived                                                                    | the continuity of the one inflow through the two fronts: a region behind one gap reads that gap's share and nothing of the fringe                                                                                                                                              | 5.7      |
| The uncertainty relation           | declaration, with its inequalities                                         | $\Delta y k \geq \pi$ from the region, $\Delta \omega \geq 2\pi/N$ from the interval, $\sqrt{Np(1-p)}$ from the draw                                                                                                                                                           | 5.3      |
| No signalling                      | derived                                                                    | $P(A+) = (\cos^2 a + r^2 \sin^2 a)/(1 + r^2)$ , independent of $b$ for parts of any amplitudes                                                                                                                                                                                 | 9.4      |
| A body's count in clicks           | derived; the unit of credit declared                                       | the share summed over the body's Nodes, what a full absorber credits; under a moving well it drifts by Eq. (13); a body's click one transition, $\hbar \omega$ per click                                                                                                       | 7.4      |
| Magnetism                          | derived under the holder's rotation (S.33); not under the plain read       | $F_C/\gamma$ : the sign's halved current over the same wall as the time level, $E_s T$ ; 0.995 at 0.1c and 0.866 at 0.5c                                                                                                                                                       | 8        |

## 5.7 The price of locality

Two consequences of the model's locality, stated against nature's clicks: exclusivity and Born's rule are the instrument's declarations (Section 5.3); and a pair is one laid record, its phase a shared classical variable inside Bell's hidden variable [24], so every local credit of it gives  $|S| \leq 2$  (THEOREM; CLICKS) [25], while the meeting through the root gives  $S = 2\sqrt{2}$  with nothing signalled, resting on the credit rule, a declaration (ASSUMPTION, FITTED; Section 9.4). A which-way measurement without absorption is no third price: nature has no detector that marks a path without a click, and the model's opaque detector in one gap loses the fringes as nature does [1, what is open, item 4; the rows against nature, (g) and (h)][3, the click is the meeting of the future with the past; go over the paper and see which mismatches with nature the clicks resolve].

## 5.8 The formulas of clicks

Table 2 collects the formulas of clicks the model gives, each with its status and the section that derives it [3, the click's algebraic definition; the mathematics of the bound bodies and how it bears on the clicks].

## 6 The families and the universe

A family is a band of the one rule and not a separate law. Its row in the universe file holds only what the rule and the geometry do not fix: its pair  $[\text{num}, \text{den}]$ , with  $\cos \omega_0 = \text{num}/\text{den}$ , and its shape: for a family of quanta its dimension, for a field its sources with its divisor. A pair written over the GameBoard's one denominator  $\Gamma$  makes a mass one integer  $m$ , with  $\cos \omega_0 = m/\Gamma$  and the wall  $6\Gamma^3$ ; a reduction to lowest terms is a reading and never an act of the loader, so such families share one resolution of their remainders [1, the families from the rule; the mass is one integer; the wall is one].

**How a family enters.** A family enters the world as one row of the universe file with two keys and nothing else: its pair, and its shape, which for a family of quanta is its dimension, 1 or 2, and for a field its sources, the form, the form with the tensions, or the Wronskian, with the field's divisor  $E_s$  and, for the massless field, its rest. Its rotation at rest, its band, its coefficients and its wall,  $w = 6 \text{den} \Gamma^2$ , follow from Eqs. (1) and (4); its parts are its sources' count, one real line per source; its reach is set by its gap (Section 6.4); its reads are its declaration's, gravity's and the binding field's at the weight 1 (Section 3.4), and it sources what it reads at the weight it reads with. Every record clicks, a field's included, since no row declares whether it clicks. Table 3 gives the rows of the universe, and Section 6.6 the numbers of the world [1, a family's declaration; the families from the rule][2, how to add a family].

### 6.1 The exact bands, and the third band

The line of one Node at rest,  $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$ , is exact in integers only where  $2 \cos \omega_0$  is an integer. So the rule carries exactly three exact massive rotations,  $2 \cos \omega_0 = 1, 0$  and  $-1$ , with the periods 6, 4 and 3 and the pairs  $[1, 2]$ ,  $[0, 1]$  and  $[-1, 2]$  (THEOREM, GAMEBOARD). A record on an exact band at rest in the vacuum leaves no remainder in its rest rotation, so it never clicks and only turns. The exact bands are exact only in the vacuum: at a Node of pace  $p_0$  the band's rest rotation is  $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2$  with  $g = (\text{den} - \text{num})/\text{den}$ , an integer at  $p_0 = \Gamma$  alone, so inside a well a record on an exact band leaves a remainder and clicks like every family's [1, the exact bands; the exact bands are exact only in the vacuum].

The bands  $[0, 1]$  and  $[-1, 2]$  are allowed by the rule and unused by the universe [1, the third and the colours, closed as absent].

### 6.2 The rule's own universe

The universe of the model is the one whose every number is the rule's own: the pairs from the integers 1, 2 and 3 of the three exact massive rotations and the three axes, every divisor 1 except gravity's, and the remaining free numbers named. Its rows stand in Table 3 [1, the families from the rule; the rule's own universe][2, the input].

The universe's integers are the Node clock  $\Gamma = 6,000$ , a multiple of 3 so that the matter pair's  $2/3$  is exact over it; the quantum's norm  $T = 2^{15}$ , the normalisation of the quadratic form; and two bounds that are not physics, the integers' width, 63 bits, and the Link unit, 16, the tension's resolution that sets the amplitude bound.  $\Gamma$  and  $T$  are units, as  $c$  and  $\hbar$  are in nature, and no line of Rule3 fixes  $T$ , since the rule is linear and the scale of its form is free; the energy line of Section 10.1 ties  $T$  to  $\Gamma$ ,  $E_s$  and  $k_w$  only where a plane family reads the holder of the sign, which no family of this universe does. Every field of the file is a record stepped by Rule3 at its pair and written by the one write; a world's starting rest is the fixed point of the division act, computed once before the first interval (ASSUMPTION) [1, the final unification][3, two units and the bounds; nothing that is not Rule3 remains].

### 6.3 What each family carries, and how gravity enters as a clock

What a family carries is set by what Rule3 reads and what a record can write, not by a name (DERIVED, GAMEBOARD). Rule3 reads two things of a Node and no third: the clock's pace  $p_0$ , one number, and the Link's paces, the three directions of the rule's vector, so a field that acts on the paces carries at most one time part and three axis parts,  $1 + 3$ . A record of one real line, neutral matter or light, has a form  $D$  and a stress and no Wronskian; a record of a plane, charged matter, has a Wronskian  $W$  as well; a field is always real, one line per source, since a plane's own  $W$  would be its own source. The shapes follow with no choice: *the scalar*, one real line, for matter, light and the binding field; *the plane* for charged matter; *one plus three*, the time part with the three tensions, for gravity, sourced by the form and by the diagonal of the record's own stress (Section 4.3); and *two real lines never summed at a Node* for the pair family of light in Bell's world. The law as it stands takes no vector part of a field and no off-diagonal stress, since Rule3 reads them nowhere; for a real record no current's read exists within the three tests, so no frame dragging of neutral bodies (Section 10.3) [1, the final unification; the tension][3, every family has a dimension].

**How gravity enters: a slowed clock, and no curved space.** There is no metric, no force and no law of motion in the model. Gravity enters in three steps. (i) *The write*: at every Node in each interval every family of quanta writes its form over the action,  $D_i \text{div } T$ , into gravity's field over the divisor  $E_g$ , by the second line of Eq. (1). (ii)

**Table 3** The rule’s own universe, its four families as the file `examples/events/rule.json` holds them (DERIVED from the two keys of a row, its pair and its shape): each family with its pair [num, den] (in brackets over  $\Gamma = 6,000$ ), what it holds at its divisor, its shape, what it reads, and its free number. A further family, the pair family of light, two real lines never summed, is declared by Bell’s world and not by this file (Section 9).

| Family                                                                                                                 | Pair                           | Holds, at the divisor                                                                                                                         | Shape                                                                                               | Reads                                                                                          | Its free number                                                        |
|------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| gravity, the time part and the three tensions                                                                          | [1, 1]<br>([6000, 6000])       | the content, $E_g$ (1,000 in the bodies’ worlds, 10 in the gravity world; $10^6$ in the three light-only worlds, gravity off in all but name) | 1 + 3 real lines, sourced by the form and the three tensions                                        | the content, its own level among it (the law’s line, implemented)                              | $E_g$ , the hierarchy; its rest $c_{vac}$ , 0 by declaration           |
| the charge, the holder of the sign, whose quanta are light, born by the write                                          | [1, 1]                         | the sign, 1                                                                                                                                   | 1 real line, sourced by the Wronskian of its readers                                                | the holders of the content, at the weight 1; never its own level                               | $k$ , the write weight, no key of the file (the plain write, $k = 1$ ) |
| the binding field, not the strong force (Section 10.3); its own events have the gap                                    | [2400, 2401]                   | the content, 1                                                                                                                                | 1 real line, sourced by the form                                                                    | the content, its own level among it (the law’s line, implemented)                              | the pair, its range $R = 20$ Links, a world’s choice                   |
| $\cos \omega = \text{num}_b / \text{den}_b$<br>matter, the scalar; a quantum that leaves a body is a message of matter | [ $m$ , den]<br>([4000, 6000]) | nothing                                                                                                                                       | dimension 1, one real line; a charged body of the same pair is a second row of dimension 2, a plane | the holders of the content, $x = (L_b + L_g) / \Gamma$ ; the plane also the holder of the sign | $m$ , the mass                                                         |

*The field:* gravity’s field, [1, 1], steps by Rule3 at its pair [1, 1] and the divisor  $E_g$ , so around a static source it rests at Poisson’s solution,  $3G(r)s$ , which is  $3s/(4\pi r)$  far away; a change travels at one Link per interval, retarded. (iii) *The read:* every family reads the field’s level  $c$  at its Node into its own coefficients, Eq. (3), the clock’s pace once and each Link’s pace twice. What nature calls curved space is here the Link’s pace, light’s index  $n = e^{2x}$ ; no space is bent in the model. The factor two on the Link is the one assumption of Section 3.4, the clock’s composition the rule’s own [1, the paces; the clock once and the Link twice][1, the same clock and the same ruler for every act; the paces compose].

## 6.4 The reach of a field, and the binding field

A field’s rest solves  $\text{num} \Delta^2 a - 6(\text{den} - \text{num})a = -3\text{den} \sigma$ , its line with the source once an interval, with  $\Delta^2$  the six-neighbour second difference. Outside a body its rest therefore falls by  $e^{-\kappa}$  per Link with  $\cosh \kappa = 3\text{den}/\text{num} - 2$ , the family’s own band at zero rotation: the gap is both the mass of the field’s quanta and the inverse reach of the field, one number (DERIVED, GAMEBOARD). [1, 1] has no gap, and its rest is Poisson’s over the whole GameBoard. [1, 2] has  $\cosh \kappa = 4$  and falls by 0.127 per Link, a contact well of  $1/\kappa = 0.48$  Links’ reach. A pair with num just under den has  $\kappa^2 = 6\text{den}/\text{num} - 6$  to second order, a range  $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$  Links, which is Yukawa’s form [47] (S.11) [1, the well].

**Assumption 4** (the binding field and gravity’s divisor, decided with a reservation). The well of a record,  $D_i \text{ div } T$ , is laid into two fields. The binding field, a family with a gapped pair  $[\text{num}_b, \text{den}_b]$ ,  $\text{num}_b < \text{den}_b$ , at the divisor 1, stepped by Rule3 at its pair as the massless field is and started at its rest, which with the gap is a sum with a range,  $R = \sqrt{\text{num}_b/(6(\text{den}_b - \text{num}_b))}$  Links and  $e^{-r/R}/r$  beyond it: it binds the bodies and is not the strong force (Section 10.3). Its own travelling events have the gap  $\cos \omega = \text{num}_b / \text{den}_b$ , a quantum with a mass whose Compton length is  $R$ , Yukawa’s own line [47] and no input. The massless field [1, 1] at the divisor  $E_g$  declared by the file: gravity, long-ranged and weak by  $E_g$ . Every family reads the holders its declaration names, at the weights it names, and sources each at the weight it reads with: gravity’s and the binding field’s levels are read by every family at the weight 1, the two fields among the readers, each its own level among the content, and the nuclear holders of Section 10.3 by the nucleon families alone (the law’s line, Section 3.4; the selective read a declaration, ASSUMPTION).

The first pair is [2400, 2401], with  $R = 20$  Links, and the field’s quantum has the rest frequency 0.029 per interval. The range is a world’s choice, tied in an earlier round to a cloud read as the proton, a reading the families replace

(Section 10.3). In the implementation's worlds  $E_g$  is declared large, 1,000 in the bodies' worlds and 10 in the gravity world, so that gravity does not enter a body's binding. Nature's  $E_g$  is read from  $G$  as the sign holder's  $k$  is from  $\alpha$ , and the hierarchy of gravity and electricity is the product of the two declared weights and the quantum's energy against its unit of charge (Section 10.1). The decision carries a reservation, that it may be wrong [3, the two rows][1, what is open, item 20].

## 6.5 The vacuum content

**Assumption 5** (the vacuum content is 0, a declaration fixed by nature's clicks). The massless field rests at a declared level everywhere, the vacuum content, the key **rest** of the universe file, derived by no line of the law; declared 0 where the light-only files carry the key and absent, hence 0, in the universe file at  $\Gamma = 6,000$ .

A uniform rest  $c_{\text{vac}}$  is a change of units (Section 3.4), which only nature's clicks can tell. Under the law's line every field reads the content as every family does, so the massless field's own travelling events have light's paces identically in every level: gravity's events and light through the same wells arrive together, the difference of their two Shapiro delays 0 and  $\gamma_{\text{GW}} = \gamma_{\text{EM}}$ , where GW170817 bounds  $\gamma_{\text{GW}} - \gamma_{\text{EM}}$  between  $-2.6 \times 10^{-7}$  and  $1.2 \times 10^{-6}$  [48] (DERIVED from the law's line; CLICKS; S.12). Inside matter the source is read through the pace, gravity gravitating (DERIVED, GAMEBOARD). With the two speeds one speed the measurement bounds no rest, and the vacuum content stays a declaration, 0 in the file (ASSUMPTION); the vacuum's write, which would give the faces a positive content, is a hypothesis of the law's documents (S.29) and not claimed here. [1, every row reads the content; what is open, item 22][3, the vacuum content is 0, a declaration fixed by nature's clicks].

## 6.6 The world's numbers, and how the universe was chosen

The universe is chosen by one principle stated before any row: one rule closes everything; the families are its inputs, each a band of the rule named by its pair; the units are a choice of scale; and the universe holds as few free numbers as the rule allows (ASSUMPTION) [3, the standing goal; not one number in the engine, and as few in the files as the law allows]. The road to them:

(i) a family is a band of the rule, and its row holds only its two keys; (ii) three exact massive rotations, the pairs [1, 2], [0, 1] and [-1, 2] (THEOREM, Section 6.1); (iii) the three of Rule3 is the three axes, so the massless holder of the content carries 1 + 3 parts and every other family is a scalar; (iv) one denominator for light and matter, their pairs written over  $\Gamma$ , a multiple of 3, the binding field's pair the one not over it; (v) the pairs from 1, 2 and 3 (ASSUMPTION): [1, 1] for gravity and the charge's field, and [2, 3], the first pair past the exact bands, for matter, nature's pair far past it (Sections 8 and 10.1); (vi) every divisor 1, under which the universe carried no free number but the mass; (vii) two numbers where nature's scale forced them, the binding field's gapped pair and gravity's divisor  $E_g$  (ASSUMPTION, with the reservation of Section 6.4); (viii) the vacuum content 0 where no write is declared (Section 6.5); (ix) the sign holder's write weight  $k$ , the plain write 1 and no key of the file, read from nature's  $\alpha$  as  $E_g$  is from  $G$  (Section 10.1).

What remains: two units,  $\Gamma$  and  $T$ ; two bounds, the integers' width and the Link unit; and five numbers of physics, matter's numerator  $m$ , the binding field's pair, gravity's weight  $E_g$ , the sign holder's write weight  $k$  and the vacuum content, and under the hypothesis of Section 7.5 one more, the vacuum write's rate, with every other divisor 1. The three light-only worlds read a copy of the file with the two holders of the content at the divisor  $10^6$ , so that light's own well stays under one level. That this universe is the universe is a decision and not a theorem (ASSUMPTION); the electron's mass over Planck's and the proton's mass over the electron's are open and not derived (Section 10.3) [1, the rule's own universe; the final unification][3, the rule's own universe is the universe; the final unification].

## 7 Bodies

### 7.1 What a body is, and how it is computed

A body is its record over its Nodes and the quanta that record carries, and never one Node: in plain words, a record held at its Nodes by the holders' wells, its count the number of its quanta, which clicks as Section 5.4 says when a quantum from outside meets its own transition. Its count is the share of its record (Section 4.4), a reading and not a state; its form  $D_i \text{ div } T$  is written into the fields it sources, and the fields enter Rule3 through the paces alone. The record is bound by the wells its form sources, by the binding field's screened sum with the range  $R$ , while gravity's  $1/r$  is weak by  $E_g$  and enters no body's binding. No law is written for a body: it is the support of a standing mode of the one rule, and its record moves by Rule3 alone. Four statements make a body (DERIVED from the line, with the decisions named): (a) the body is the fixed point of its binding field, standing where the sources return themselves; (b) a field's level is the rest the start solves from the bodies' forms, after which Rule3 steps it; (c) a body's quanta are its record's share, carried wherever its record's current carries it; (d) a free packet or a cloud falls as  $a = \Delta |d\omega_b/dc| \nabla c$  [1, what a body is; the four lines of the body][3, what a body is].

**How a body is computed.** The held rows' source is the record's form  $D_i \text{ div } T$ , the law's line; the implementation's start laid the count, and the generator's lay under a level is being checked, the write's factor  $N^4$  read in the field and no physical return (the open list; GAMEBOARD). By the rule's own iteration and no other formula: start from a spread of the declared count over the Nodes; write the wells that spread lays into the fields and let them rest; find the record's top mode in those paces by Rule3's own read act, iterated; scale its two levels so that the share carries the declared count; write the wells that record lays; and repeat until the levels repeat, in integers, at the fixed point of the division act. The branch is chosen by the start's width alone: a wide start reaches the cloud and a narrow one the compact body, two fixed points at one count (the generator's readings of the earlier rounds; no standing body in this round). [1, the generator; the four lines of the body][3, the generator is Rule3].

## 7.2 Stability

Let a body of  $M$  quanta have the record  $\phi$  over its region,  $\sum \phi_i^2 = 1$ , with the counts  $c_i = M\phi_i^2$ . Let every field  $r$  it sources be a hollow ( $\sigma_r = +1$ ) or a hill ( $\sigma_r = -1$ ), with the static kernel  $\mathbf{G}_r$  of its line and the divisor  $E_r$ , so that the well at a Node is  $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$ . To first order in  $\Phi/\Gamma$  the well raises the one-Node rotation  $2 \cos \omega$  by  $4g\Phi/\Gamma$ ,  $g$  the gap of Section 6.1, and the body's fixed point is a critical point on the sphere of

$$F[\phi] = \frac{\text{num}}{3 \text{ den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2, \quad (12)$$

the band's pressure less the wells' energy, each well counted once because the record is its own source (Hartree's form). The record is the top mode of Rule3's symmetric form at the paces, and  $dF/d\phi = 2\lambda\phi$  on the sphere is the fixed point's own equation with  $2 \cos \omega_b = 2 \cos \omega_0 - \lambda$  (THEOREM, GAMEBOARD; S.13). Let  $\phi_R(x) = R^{-d/2} \phi(x/R)$  be the body dilated, with  $d$  the GameBoard's dimension; the pressure is  $K/R^2$  and each field's well energy is  $W_r/R^{s_r}$ , with  $s_r$  the field's scaling power at the body's size:  $s_r = d$  beyond the field's reach and  $s_r = d - 2$  within it or for the massless field [1, the functional and the dilation].

**Theorem 4** (the stable body, under the dilation and to first order in the well). *At a fixed point of the binding row of width  $R$  the virial  $2K = \sum_r \sigma_r s_r W_r$  holds, and the second variation of  $F$  along the dilation is  $R^2 F''(R) = \sum_r \sigma_r s_r (2 - s_r) W_r$ . A body is stable only where this sum is positive; it is a minimum of  $F$  where further the Hessian on the modes orthogonal to the dilation and the translations is positive.*

(Proved in S.48, by Derrick's scaling [49] and Pohozaev's identity [50] on the GameBoard.) The theorem is a statement about the functional of Eq. (12), to the first order in the well, along the one direction of the dilation, with the dilation taken as continuous; it is not a theorem of the integer step on the full lattice, and the Hessian beyond the dilation is open (Section 10.3).

The factor  $s(2 - s)$  is  $+1$  at  $s = 1$ ,  $0$  at  $s = 2$  and  $-3$  at  $s = 3$ : a hollow holds the body where its well falls more slowly than the pressure and breaks it where it falls faster. Three consequences follow. (a) One hollow beyond its reach binds every mass on a chain and in a box has one critical point, a saddle: the body disperses beyond it and collapses within it. (b) A massless field holds the body: Poisson's well in a box has  $s = 1$ , so  $F = a/R^2 - c/R$  has one critical point,  $R = 2a/c$ , a minimum for every mass, which is Pekar's body [51]. Beside it a hollow beyond its reach adds a barrier,  $F = a/R^2 - b/R^3 - c/R$  on a Gaussian body, with the wide critical point a minimum and the narrow one a saddle, and none where  $3bc > a^2$ , the collapse, computed on the pair [5760, 6000] (COMPUTED, GAMEBOARD). (c) A collapse ends in a frozen clock: beyond first order the well saturates and the Link's pace weakens the pressure, and since no pace reaches 0 before the rounding a collapse ends in a cluster of Nodes whose clocks are frozen to the unit and never in a closed Link [1, the functional and the dilation][1, what is open, item 29].

## 7.3 The two branches, and the status of a standing body

The binding row's fixed point has two branches, a wide cloud and a compact pixel, chosen by the start's width alone (Section 7.2; DERIVED for the functional, S.13). No standing body is computed in this round: the look of the body in a uniform level (the open list) finds the generator's bodies bound but not standing within rounding over 4,000 intervals, the neutral body breathing and drifting by 7 to 10 percent of its laid share and the charged body growing under its own sign holder by 53 to 70 percent (a finding of the implementation's blind in three dimensions; GAMEBOARD). Three problems of the computation are named, none in Rule3. (1) The lay is not the integer fixed point: the generator lays the count from the continuum forms with the vacuum's kernel, and in a level the well comes out  $N^4 = 0.81$  of the laid depth for the sign and  $N^6 = 0.74$  for the content, a well 19 to 26 percent deeper than the body finds, so it relaxes and breathes. (2) The turn's convention: the engine on **main** turns both levels by the level at now, whose booked Wronskian moves by  $\mp \cot \omega \Delta \theta$  under a changing level; a breathing charged body's changes rectify and its charge grows; the time-Link turn, exact for any level, removes it and is not yet merged. (3) The rounding walk: the

neighbours read the integer level and not the carried remainder, a noise below one level per Node per interval, walking over  $n$  intervals as  $\sqrt{n}$  levels against the amplitude  $A = \sqrt{T c_i / (2 \sin \omega)}$  levels per Node,  $c_i$  the quanta per Node, 852 for the blind's body at  $T = 2^{15}$ : about 19 levels over the 4,000 intervals, 4.5 percent of the share, the size of the neutral body's drift. The fix proposed, the engine's to-do and nothing of the law: the lay at the integer fixed point, the start iterated under the body's own paces until the levels repeat in integers; the time-Link turn merged, the charged body's growth its gate; the resolution raised, the walk falling as  $1/\sqrt{T}$ , about one percent of the share at  $T = 2^{19}$ ; a standing body then stands within the walk's bound  $\sqrt{n}/A$ , a stated criterion. Until then the body's rows carry only what is computed, its definition (Section 7.1), the static fixed point and the modes by hand (S.46, S.60), the stability conditions as derived (Section 7.2) and its clicks (Section 7.4), and no sentence that a body stands under Rule3; the clouds of the earlier rounds and their numbers are the law's document's and not this paper's.

**The atom: known from clicks, laid on the GameBoard.** The atom is known from clicks, its spectra, its lifetimes and its scattering; nothing of it is seen on the GameBoard. What is tested is whether the way from the clicks to the GameBoard and back carries the atom, a supporting finding with its state named (S.46, S.52). An atom is a bound body of the holder of the sign: a nucleus, one quantum of its own family, writes the time angle  $\theta_0 = Z\alpha c/r$ ,  $c = 1/\sqrt{3}$ , into the sign holder's row by its Wronskian, electrons, quanta of the matter family, read that angle into their rotation, and the standing records Rule3 makes in that angle are the atom's levels, Schrödinger's in the slow limit. With the nucleus's angle alone they have Bohr's form,  $E_n \propto 1/n^2$  with  $a_0 = \sqrt{3}/(m^* Z\alpha)$ ; the record also reads its own level, so the form is the law's at large  $Z$  and hydrogen itself is not reproduced, an explicit finding; the lines are beats of two levels and the mass defect the click layer's reading of a frequency; the orbital Zeeman splitting has  $g = 1$  (supporting findings with their numbers in S.46; CLICKS for the levels' ratios, GAMEBOARD for the records). The electron and the nucleons are quanta of their families: an electron is one quantum of count  $\sin \omega_e$ , ten orders below the compact threshold, and nothing smaller than a quantum clicks; mass is the family's gap,  $T \sin \omega_s$ , bound or free; what decays is a body, a quantum parted from it by a click at its front, and a quantum never breaks, having no part the click credits (DERIVED; CLICKS). The criterion, one line: what nature splits with energy is a body on the GameBoard, what no energy splits is a quantum of a family; the nucleus, the atom and the molecule are bodies by the one mechanism the law has for a body, a record standing in the well of a holder, the sign holder for the atom and the nuclear holder for the nucleus (ASSUMPTION) [1, the bound body is one Node; what nature splits is a body; the nucleus][3, the electron and the nucleons are quanta, not pixels; no mass without a bound body; a bound body that breaks up is an atom that breaks up]. A nuclear holder the electron reads is excluded by the spectrum, hydrogen's  $1S$  and muonic hydrogen (DERIVED, one derivation; CLICKS; S.46); so the nuclear holders of Section 10.3 are read by the nucleon families and not by the electron, a selective read declared. S.60's table sets the atom's clicks, the lines and the thresholds, against nature's under the two lines decided for the law, with how each was derived, a supporting finding and no more, the engine's gate pending; what the atom of the model does not give is listed with its roots in Section 10.3 and S.52 [1, the atom is a bound body of the holder of the sign].

## 7.4 A body's clicks

What a detector reads of a body follows from the body's record at its Nodes, the readings of Section 4.3 and the paces, with no further line; every formula below is a derivation with its status and its fence, and none is a run's result [3, the mathematics of the bound bodies and how it bears on the clicks].

**A steady rotation writes no light.** A body at rest with the plane record  $\text{re} = A \cos \omega_b t$  and  $\text{im} = eA \sin \omega_b t$ ,  $e$  its ellipticity, has the Wronskian  $W = -eA^2 \sin \omega_b$  at every interval, so the holder of the sign's source is constant and its field stands at its rest: no travelling events, no click. Rule3 keeps a body's total Wronskian, and an isolated Node's exactly, so a one-Node source cannot breathe and a body's breathing moves  $W$  between its Nodes; its light is written by the second moment of its charge, the monopole constant by the continuity identity, so the light born carries the breathing's frequency and not the rotation, and a far detector's click rate is that current's share through its front (S.42). So a symmetric body at rest clicks nowhere, and a body is seen by what it absorbs and by what it writes when it breathes or moves (DERIVED; the Wronskian and the source GAMEBOARD, the far detector's rate CLICKS) [1, light is born by the write].

**The invariant, and the count's drift.** For one Node of a body,  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$  with  $\omega$  changing slowly keeps  $A^2 \sin \omega_b$  invariant (S.14), the discrete adiabatic invariant of Rule3's line, while the form  $D = A^2 \sin^2 \omega_b$  and the share are not; the recurrence stepped over 20,000 intervals keeps  $D/\sin \omega$  to  $4 \times 10^{-5}$ . The share in quanta at a Node,  $(\Gamma^2/p_i^2) A^2 \sin^2 \omega_b/T$  (Section 4.4), therefore moves under a moving well as

$$\frac{N(t_2)}{N(t_1)} = \frac{\sin \omega_b(t_2)}{\sin \omega_b(t_1)} \cdot \frac{p_i^2(t_1)}{p_i^2(t_2)}, \quad (13)$$

with no current through any Link: the first factor is the rotation's own change under the clock's pace, the second the Node's weight in the share (DERIVED for the invariant, to the first order in the slow variation, and for the drift; GAMEBOARD; S.14). The invariant is, in the model, what nature calls the number of quanta,  $E/\omega$ , and the share is nature's energy; the same identity is Planck's relation on the lattice, share =  $I \sin \omega$  with  $I = A^2 \sin \omega / T$ , so that  $E = \hbar \omega$  holds with  $T$  for  $\hbar$  and  $\sin \omega$  in place of  $\omega$  (DERIVED; CLICKS) [1, the count is the record's share; the paces' own change moves the share with no current].

**The clock's click rate.** A bound body's rotation reads the content once: at rest  $\cos \omega_0(U) = \cos \omega_0 + 2gU$  to first order, so its rate falls by  $f(\omega_0)U$  with  $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ , 1.063 at [2, 3] and 1 as the gap closes,  $f = 1 + \omega_0^2/12 + \dots$ , the finite-rotation correction in  $\omega$  of a rate that is exactly  $N$  in  $2 \sin(\omega/2)$  (S.31). A light clock between two faces falls by  $2U$ , the Link's pace read twice (Section 10.2; S.15) (DERIVED, CLICKS).

**The resonance.** A bound record at the rotation  $\omega_b$  reading a light level into its content is modulated by  $\epsilon = 4g(1 - 2x_0)b_L/\Gamma$ ; the resonant part at  $\omega_L = 2\omega_b$  grows the amplitude at  $\mu = \epsilon/(4 \sin \omega_b)$  per interval inside the full width  $\epsilon/\sin \omega_b$ , and a body with two bound modes  $\omega_i < \omega_j$  transfers between them at the difference  $\omega_j - \omega_i$  at Rabi's rate  $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin \omega_i \sin \omega_j})$  (DERIVED, two derivations; S.16). Under light's declared read, the holder's rotation (Section 5.5), the Link phases couple the modes through the inter-mode current, resonant at the difference line for drives below  $\omega_j + \omega_m$ , Rabi's rate  $\Omega_R = \theta_L \omega_L |d|/2$  for a small line, oscillatory at every detuning with the envelope  $\Omega_R^2/(\Omega_R^2 + \Delta^2/4)$ ; at the sum  $\omega_j + \omega_m$  the same coupling drives the two senses together, a pair line, outside this section's absorption and emission (the atom's lines at optical frequencies lie far below it); the one-mode line at  $2\omega_b$  has no analogue there (DERIVED; S.43). The read's side is computed; the fraction absorbed, with the body's write back, is open (GAMEBOARD for the record, CLICKS for the body's share) [1, the rows against nature, (y)][1, the two-mode line].

**A body's count in clicks.** A body's count is its share summed over its Nodes, what a full absorber credits, and under a moving well it drifts by Eq. (13) (DERIVED, CLICKS). The implementation's declared region credits the share, a bolometer's reading, Millikan's slope 0. Nature's detector is a body, and the model declares its click as one quantum's change in the absorbing body, the detector's fourth declaration beside its window, its region and its floor (ASSUMPTION) [3, the whole of a click is the detector's own transition]: the absorber answers at its own rotation by the two-mode line above (S.43), one transition per click, taking the share  $T(\sin \omega_j - \sin \omega_i)$ , one photon's share  $T \sin \omega_L$  to the order  $\omega_i \omega_j$ , below  $10^{-18}$  at nature's gap. So a body counts photons and reads  $\hbar \omega$  per click, Millikan's slope  $T$  (DERIVED; S.14); The share stays the conserved form and the draw's measure, so every ratio of the checks is unchanged (THEOREM); the implementation credits the share, the declared unit not yet implemented (CLICKS) [1, the count is the record's share].

## 7.5 Clusters: how they enter, and what each thing represents

**How a cluster enters.** The model writes no line for a cluster. One assumption names the universe it describes: every body is bound, no body is free, and a record that is in no body is a message between two clicks (ASSUMPTION). A cluster is bodies standing in one another's wells; between two bodies there is nothing but the click, the light of the holder of the sign and the two real fields [1, the bound body is one Node; the universe is bound; the algebra of clusters].

**The clicks between bodies are clicks in the one sense.** A body is a region of Nodes with a front, the Ports where its tail meets its neighbour's, so a body is an instrument in the meeting's form: the click between two bodies is the pair  $(R, t)$ , one whole quantum credited to the region of the taker at the interval  $t$ , drawn by the share  $Q_R/W_c$  of the inflow through its front on the forward orbit. A body's count is its share summed over its Nodes, and it moves only by the clicks at its front, up to the drift of Eq. (13) (DERIVED from the definitions; CLICKS) [3, the click's algebraic definition; the click is the meeting of the future with the past].

**What each thing represents in a cluster.** The three levels at a Node are the nesting: the binding field's level is the matter's tail, the sign holder's the charge's well and the massless field's gravity's well, three kinds of binding by rank, quanta into a body, bodies into a charged cluster, charged clusters into a gravitational cluster (DERIVED, GAMEBOARD); the couplings are at the edge, every divisor 1 but gravity's  $E_g$ , the universe of clusters carrying  $\Gamma$ ,  $T$  and the two holders' weights and nothing else (ASSUMPTION for the divisors) [1, the levels at a Node are the nesting; three kinds of binding, not three levels; the hierarchy is recursive; the click is the pointer; the observer is a cluster; the couplings are at the edge][3, what a body is; the rule's own universe is the universe].

**The clusters' cosmology.** The universe of the model is a finite ball of bodies flying through the static lattice, every body pulled by the others' inverse square at Kepler's  $G_K$ , Newton's constant as the orbits read it (Eq. (20); Section 8,

(j)). A shell of dust at the radius  $a$  feels the mass inside it, so  $\ddot{a} = -(4\pi G_K/3)\rho a$  and  $\dot{a}^2/a^2 = (8\pi G_K/3)\rho - K/a^2$ , Friedmann's dust equations exactly for the ball's interior [52],  $K > 0$  where the ball is bound; a receding emitter's redshift is  $1 + z = 1 + \beta$  at first order, Doppler's and Hubble's known forms derived here, where the two derivations, the Doppler period of row (m) times the moving clock of row (e), agree exactly (DERIVED, two derivations; CLICKS; S.29). The paper's cosmology is claimed at  $z \ll 1$  and not beyond; the second order, where the deceleration is read, is named among the findings of Section 10.3 [1, the clusters' cosmology].

## 8 The forms of nature the rule arrives at

Each form below follows from the rule's coefficients, read by the geometric optics of a packet's band, before any run; the form of nature beside it is never an input, the one assumption of Section 3.4 excepted. The coefficients contain no metric, no Lorentz transformation, no potential and no force, only the paces of Section 3.4, the clock composing by the rule and the Link the clock twice by the one assumption.  $U = c/\Gamma$  denotes the model's potential at the point named, the clock's share per level, and  $U_b$  its value at the impact parameter  $b$  [1, the rows against nature].

**Two potentials.** The forms are stated in two potentials. The model's own potential is the clock's share,  $U = c/\Gamma$ , the content  $x$  that the paces read; no instrument reads it. Every test of gravity takes  $GM$  from Kepler's law of the orbits of matter and  $c$  from light, so it reads  $U_K = GM_K/(c^2 r)$ . In the model a free packet of matter of the pair [num, den] falls in a content gradient at  $a = c^2 [2 \cos \omega_0 / (1 + \cos \omega_0)] |\nabla x|$ , exactly from its band, Eq. (4) (S.18), so the orbits of matter measure

$$U_K = \frac{2 \cos \omega_0}{1 + \cos \omega_0} U, \quad 0.800 \text{ at } [2, 3], \quad 1 \text{ as the gap closes}. \quad (14)$$

Each form below, lettered as the law's rows are, is written first in  $U$ , where the composed paces give  $\gamma = \beta = 1$  exactly in the post-Newtonian words, the metric  $N^2 = e^{-2U}$  and  $h^2 = e^{2U}$ , and the redshift's factor  $f(\omega_0) = 1 + \alpha_U$ ,  $\alpha_U = 0.063$  at [2, 3], and then in  $U_K$ , where the same algebra gives  $\gamma_K = \text{den}/\text{num}$ ,  $\beta_K = (\text{den} + \text{num})/(2\text{num})$  and  $1 + \alpha_K = \tan \omega_0 / \omega_0$ : 1.5, 1.25 (the small-gap closed form under the quadratic band; the lattice band's advance in (d)) and 1.329 at [2, 3], and 1 as the gap closes. The two columns are one algebra, and nature's tests read the second (DERIVED, CLICKS; S.18, S.19) [1, the rows against nature][3, general relativity in the law].

- (a) *The redshift* (DERIVED, CLICKS). Two clocks of one family at the levels  $c_1$  and  $c_2$  tick in the ratio  $1 - (U_1 - U_2)$  to first order; the rest rotation of a bound record, the clock this row names, shifts by  $f(\omega_0) U$  (Section 7.4), 1.063 at [2, 3] and 1 as the gap closes. Against the Newtonian potential the orbits measure it is  $z = (1 + \alpha_K) U_K$ , 1.329 at [2, 3], where nature's clocks give  $|\alpha| < 10^{-4}$  [38]. Beside it: Pound and Rebka's line [56][38].
- (b) *The bending of light* (DERIVED, CLICKS). At a Node of content  $x$  light's index is  $n = e^{2x}$ ,  $1 + 2x$  to first order, because the Link's pace reads the content twice. Far from a body its field is the massless field's rest,  $L(r) = 3\sigma_{\text{tot}}/(4\pi r)$ , with  $\sigma_{\text{tot}}$  the body's wells per interval over the field's divisor  $E_g$ , and a beam passing at the closest distance  $b$  turns by the transverse gradient of the index integrated along the path (S.20):

$$\alpha = \int_{-\infty}^{\infty} \frac{\partial n}{\partial b} dl = 4U_b, \quad U_b = \frac{3\sigma_{\text{tot}}}{4\pi \Gamma b}, \quad (15)$$

twice the angle  $2U_b$  that a Newtonian pull of the same potential gives. The factor two is the ruler going with the clock,  $hN = 1$ , the one assumption of Section 3.4. Against the Newtonian potential the orbits measure the bending is  $(1 + \gamma_K) 2GM_K/(c^2 b)$  with  $\gamma_K = \text{den}/\text{num}$ , 1.5 at [2, 3], where Cassini's tracking gives  $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$  [38][57]. Beside it: Einstein's  $4GM/(c^2 b)$  [38]. The bending is the  $\beta = 1$  case of the push on a moving record:

$$\mathbf{F} = \frac{E^2 + P^2}{E} \nabla U = E(1 + \beta^2) \nabla U, \quad \beta = \frac{P}{E} = \frac{v}{c_{\text{local}}} \quad (16)$$

with  $E = \omega$ ,  $P = cN^2 k$  the momentum at the composed paces and  $\beta$  the record's speed over the local light speed  $cN^2$ : at rest  $E \nabla U$ , for light  $2E \nabla U$ , Eq. (15) its  $\beta = 1$  case, its bending exact in the gap, while the orbit's second order at the computed pair carries the band's  $k^4$  term (S.19) (DERIVED from Eqs. (3) and (4) under the one assumption, fitted; S.58) [1, the push on a moving record]. The paper's momentum is the wave number, so the lattice form is  $(E^2 + c^2 \mathbf{p} \cdot \mathbf{p})/E$  and  $E^2 + 3\gamma \mathbf{p} \cdot \mathbf{p}$  only for  $\mathbf{p} = E\mathbf{v}$ ; at the computed pair the coefficient of  $\beta^2$  is  $\gamma_K - \alpha_K/2 = 1.335$  and the rest push  $f(\omega_0) E$ . As physics it is derived here and coincides with the known post-Newtonian deflection factor  $1 + \gamma\beta^2$  at  $\gamma = 1$ , derived and not declared; nature is met by its click formula, the deflection angle's ratio  $1 + \beta^2$  to Newton's (CLICKS).

- (c) *The delay of light* (DERIVED, CLICKS; S.20). Light's coordinate speed at the content  $U$  is  $e^{-2U}/\sqrt{3}$ , so its index is  $1 + 2U$  to first order, and a signal grazing a body at the impact parameter  $b$  from  $x_1$  to  $x_2$ , with  $U = \mu/r$ , is late by  $2\sqrt{3}\mu \ln(4x_1x_2/b^2)$  intervals: Shapiro's delay with  $\gamma = 1$ . Beside it: Shapiro's delay [58].
- (d) *The perihelion* (DERIVED, CLICKS). A slow body's orbit in the composed clock's metric,  $N^2 = e^{-2U}$  and  $h^2 = e^{2U}$ , is the post-Newtonian orbit with  $\beta = \gamma = 1$  in  $U$  for the quadratic band, the small-gap limit, so the periapsis advances by  $6\pi\mu/(a(1 - e^2))$  per orbit with  $U = \mu/r$  and  $e$  the eccentricity, Einstein's value, Mercury's 43 arcseconds per century at nature's pair; at the computed pair the band's  $k^4$  term, of the same order as the relativistic one, adds of the order of ten percent to the advance, orientation dependent, vanishing as  $\tan^2 \omega_0$  (DERIVED at the small gap; the quadratic band an ASSUMPTION at the computed pair; S.19, S.34). Against the Newtonian potential the orbits measure the advance is  $(2 - \beta_K + 2\gamma_K)/3$  of Einstein's,  $\beta_K = (\text{den} + \text{num})/(2\text{num})$ , 1.25 at [2, 3], the small-gap closed form under the quadratic band (with the lattice band the advance gains of the order of ten percent, orientation dependent, and no single  $\beta_K$  stands for it), where the planets' and the Moon's orbits give  $|\beta - 1| < 10^{-4}$  [38]. Beside it: Einstein's  $6\pi GM/(c^2 a(1 - e^2))$  [38].
- (e) *The moving clock: the band's kinetic scale and the fourth order* (DERIVED, CLICKS; S.21). The tick's factor of a packet moving at the velocity  $v$  is  $f = \Omega(k)/\omega_0$ ,  $\Omega(k) = \omega(k) - k\omega'(k)$  the rotation at the moving centre; from Rule3's exact band,  $f = 1 - v^2/(2c_m^2) - (2\omega_0/\sin 2\omega_0)v^4/(8c_m^4) + O(v^6)$  with  $c_m^2 = \omega_0/(3\tan \omega_0)$  the band's kinetic scale, the fourth order's coefficient 1.6926 at [2, 3] and 1 as the gap closes. The kinetic scale  $c_m^2/c^2 = \omega_0/\tan \omega_0$  ( $1 - \omega_0^2/3$  at a small gap) is 0.752 at [2, 3], so  $c_m$  is 13 percent below light's, and the band's largest group speed  $2/3$  of light's (S.21); at nature's matter pair  $c_m$  lies below light's by  $\omega_e^2/6$ ,  $2.3 \times 10^{-20}$ , and the largest group speed by  $\omega_e^2/2$ ,  $6.8 \times 10^{-20}$ , at the LHAASO bound,  $8 \times 10^{-29}$  and  $2.4 \times 10^{-28}$  at the axis bound (Section 10.1), the electron subluminal, where nature bounds the difference of matter's and light's limiting speeds with its sign: the subluminal case at about  $10^{-20}$  by the Crab's synchrotron [61], the superluminal at  $5 \times 10^{-21}$  [62], the earlier bounds  $10^{-15}$  to  $10^{-20}$  [59]; the subluminal bound, where it holds, binds  $\omega_e$  near  $10^{-10}$ . Beside it: Lorentz's factor, reached as  $c_m \rightarrow c$  [3, the click].
- (h) *De Broglie's relation* (DERIVED; GAMEBOARD for the band, CLICKS for the fringe spacing that reads it; S.22). A free record of a matter family has the band  $\cos \omega_b = (\text{num}/(3\text{den}))(2 + \cos k)$ , the group velocity  $v = (\text{num}/(3\text{den})) \sin k / \sin \omega_b$ , and to second order  $k = \omega_b v / c_m^2$  with  $c_m$  the band's kinetic scale of (e). Beside it:  $\lambda = h/p$  with the rest rotation as the mass.
- (i) *The fall* (DERIVED, CLICKS). In a pace gradient a free packet or a cloud accelerates at  $a = \Delta |d\omega_0/dc| (c_+ - c_-)/D$  toward the well,  $\Delta$  its mode's top velocity and  $c_\pm$  the level at its two faces  $D$  Links apart. So bodies with the same  $\Delta d\omega_0/dc$  fall alike. Beside it: Galileo's fall [38]; the binding by the holders of the content gravitates as in nature, two bodies of one count and different binding falling alike, while the seat of the electromagnetic binding energy is open (DERIVED; CLICKS; S.38). Between families: a free packet of the pair [num, den] falls at  $c^2(1 - \tan^2(\omega_s/2)) \nabla U$  exactly, so two families of different gaps fall apart by  $(\omega_n^2 - \omega_p^2)/4$  at small gaps; with the proton and the neutron two families, beryllium against titanium gives  $\eta = 32.3\omega_e^2$ , below  $4.4 \times 10^{-18}$  at the LHAASO bound, under nature's  $10^{-13}$  for that pair [38] and  $10^{-15}$  for titanium against platinum [37] (DERIVED, CLICKS; S.47) [1, the rows against nature, (i)].
- (j) *Newton's pull and orbits* (DERIVED, CLICKS). The massless field rests at Poisson's solution with the sources and at Laplace's outside a body,  $c(r) = 3s/(4\pi r)$ , so the acceleration falls as  $1/r^2$  and a circular orbit's  $v^2$  as  $1/r$ . Beside it: Newton's pull and Kepler's fall-off.
- (m) *A free record keeps its wave number* (THEOREM, GAMEBOARD). Rule3 is invariant under translation, so a free record keeps its wave number and rotation exactly, and a radiating body's period is read unchanged at every distance. Beside it: Hubble's law as Doppler clicks of matter flying apart through the static lattice, with no stretching of space (Section 7.5).
- (p) *The lens: a content bends light and matter alike, by one formula* (DERIVED, two derivations; GAMEBOARD for the index, CLICKS for the focus and the clicks; S.28). A stationary boundary conserves the rotation, so for every family a record of wave number  $k$  entering a content  $U$  takes the wave number  $k_{\text{in}}$  given by

$$1 - \cos k_{\text{in}} = \frac{1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}}{N^4}, \quad N = e^{-U} \quad (17)$$

light's index  $k_{\text{in}}/k \rightarrow e^{2U}$  at long wavelength (nature's  $n = 1 + 2\Phi/c^2$ ), exact in the law's own variable as  $\sin(k_{\text{in}}/2) = \sin(k/2)/N^2$ , with the lattice's chromatic term  $k^2(e^{6U} - e^{2U})/24$  at finite  $k$ ; slow matter's  $n_m^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num} k^2)]/N^4$ , growing as  $1/k^2$  (nature's  $n^2 = 1 + 2\Phi/v^2$ , with the Kepler column's ratio at the computed pair, Eq. (14)). *What it does:* a slab of content bends every record toward its thick side, light a little and slow matter much; the fall of a body is the refraction of its own record; and a window of content thick in the middle is a lens, gathering the record's share at a focus. *One quantum through it* (Section 5.4): the whole record passes the lens and its share gathers at the focal spot; the one quantum clicks once,

at one Node of the spot drawn by the detector with its seed over the spot's shares and written there, the levels that arrived taken, the count down by one; the next quantum draws again; the spot is the clicks' accumulation, one per quantum, the image built click by click, as nature's single-photon images are (the formula derived from Eq. (4) at the composed paces; the one click per quantum the credit's declaration; the write at the one Node the instrument's declaration, its engine round pending) [1, the content's index for every family; the rows against nature, (p)][3, one quantum through a lens, a beam splitter and an interferometer is the law's own].

**What goes in and what comes out.** The model puts in one rule, the paces' read under the one assumption of Section 3.4 and the run files' numbers. In the model's own potential the bending (b), the delay (c) and the perihelion (d) are Einstein's exactly,  $\beta = \gamma = 1$  in that column, light's ray metric; the post-Newtonian column at the computed pair departs with the gap. At the rule's own matter pair the model therefore does not describe the gravity of the solar system; the closing gap is nature's pair, where  $\gamma_K - 1$  is below  $10^{-19}$  at the LHAASO bound and  $2.4 \times 10^{-28}$  at the axis bound (Section 10.1), and whether the model's bodies stand at that pair is not shown. Read as a metric, the composed clock gives  $N^2 = e^{-2U}$  and the ruler  $h^2 = e^{2U}$ , against the isotropic Schwarzschild chart's  $1 - 2U + 2U^2$  and  $1 + 2U + 1.5U^2$ , the spatial second order 2 against 1.5. One click of the strong field reads the difference: light's index  $e^{2U}$  places the photon sphere at  $r = 2m$  with  $U = m/r$  (S.30), so the capture radius of a black hole's shadow is  $b = 2em = 5.44m$  against Schwarzschild's  $3\sqrt{3}m = 5.20m$ , 4.6 percent larger, an explicit prediction (DERIVED under the one assumption, which is fitted; CLICKS; S.30). Beside it: the Event Horizon Telescope's shadows, Sgr A\*'s fractional deviation from Schwarzschild  $-0.08 \pm 0.09$  [63], 1.4 standard deviations from the prediction under the VLTI prior (1.0 under Keck's), and M87\*'s  $-0.01 \pm 0.17$  within 0.3 [64]. Rule3's step times the record's conjugate gives an exact continuity identity: the sign's density and its halved Link current are one conserved four-current, and the time turn with the three Link phases is one covector by minimal coupling; so the odd lines are sourced by the halved current over the same wall as the time level, no number matched, and a co-moving reader feels Coulomb's force over  $\gamma$  (DERIVED; CLICKS; S.33). A source and a reader of one family see one light speed, the band's own ( $c_s^2 = \text{num}_s / (3 \text{den}_s)$ ) in the variable  $2 \sin(\omega/2)$  and  $c_m^2$  in  $\omega$ , one as the gap closes;  $1/\gamma_s$  uses  $c_s$ , light's  $c$  the same to  $10^{-20}$  in nature at the LHAASO bound [1, the odd line's source is the sign's own current; the rows against nature, (b2)].

## 9 The two slits and Bell: the one computed world and the check, derived and read by the implementation's checks

The paper carries no experiment against nature: the results of the worlds below are derived from the rule and read by the implementation's checks (Section 3.7). The two slits is the one computed world that stands as an experiment of the method; Bell's world is a check and not an experiment: its number is fixed by the declared credit and the parts' determinism. Nothing in a world is built to bring the right result [3, the paper; one experiment and one gate; nothing in an experiment brings the right result by construction].

### 9.1 The two slits: the world

The emitter is a declared packet of light. Its wavelength is 8 Links along  $+x$ ,  $k = \pi/4$ . Its levels are a raised-cosine packet of about ten quanta per Node,  $b = 1,328$ , over  $x = 4$  to 16, flat across  $y = 4$  to 44. The wall is the inner face at  $x = 20$  with two gaps of three Nodes,  $y = 17$  to 19 and 29 to 31,  $d = 12$  Links apart. The GameBoard is flat,  $45 \times 48 \times 1$  with the third axis folded, its  $y$  faces open and its  $x$  faces receding; the screen is the last column,  $x = 44$ ,  $L = 24$  Links from the wall. The screen is twelve declared detectors of four rows each; the run is 130 intervals, read over the intervals 45 to 130. A record uniform along the folded axis has the three-dimensional band at  $k_z = 0$ , so the flat GameBoard carries the world's band exactly (DERIVED, GAMEBOARD). On this world the law's line gives 278.0 quanta through the screen in the physical labels (277.8 in the implementation's labels), the peak of the screen's inflow at the interval 68.0 against 61.8 on the same world with the lattice refined fourfold (COMPUTED, CLICKS) [1, the rows against nature, (g)][3, the two slits; one experiment and one gate].

### 9.2 The two slits: what the model says it must give

Three readings of the one world: the known forms' prediction, Huygens's sum and Young's spacing applied to the declared world's geometry and written before the run, and the law's real line computed beside it. (a) *The fringe.* The far field's spacing is  $\lambda L/d = 16$  Links; at  $L = 2d$  the fringe is the near field's, with the minima at the rows 15 and 33 (DERIVED, CLICKS; S.23). Light's line at the vacuum's paces, stepped in real arithmetic on the declared world, gives 278.0 quanta through the screen (Section 9.1), per region as the script's table lists (COMPUTED, CLICKS; the script of Section 3.7); the Huygens sum of the two gaps gives 273 with the same central fringe. (b) *The clicks.* The screen's shares rounded to whole quanta are the expectation, and the row drawn by the shares at the declared seed is the clicks, with the draw's scatter  $\sqrt{Np(1-p)}$  per region; on the Huygens row at  $N = 273$  the central maximum

stands above its two minima by 5.7 times its scatter. (c) *The lattice's own number*, the inflow peak's 68.0 against 61.8 (Section 9.1), which the click line resolves (COMPUTED, CLICKS). [1, the rows against nature, (g)][3, the click is the meeting of the future with the past].

### 9.3 Bell: the world

The entanglement the model computes is one event with two trees: a pair is two records from one root; each is read in two directions, forward the future and backward the past; a click is where one record's two readings meet, and the pair's correlation is the partner's record read through the same root, re-weighted by what was realised. The pair is laid as the pair family, two real lines of light never summed at a Node, with equal parts. Each side's detector, a region at least half a wavelength across, reports the two parts' signed level sums over its region and knows nothing of angles; the ports enter at the credit. The reader applies the declared setting, a pair of integers  $(p, q)$ , with the port coefficients  $e(+) = (p, q)$  and  $e(-) = (-q, p)$ , exactly orthogonal, so each side's marginal is one half for any setting. The settings of the Clauser-Horne-Shimony-Holt test are  $(1, 0)$ ,  $(1, 1)$ ,  $(12, 5)$  and  $(5, 12)$ , at 0, 45, 22.62 and 67.38 degrees. The credit is the one form of the click: the joint share pairs the parts by their label across the two sides through the root,

$$J(p_A, p_B) = \sum_{\text{window}} \left| \sum_k c_k e_k(p_A) \alpha_k(p_A) e_k(p_B) \beta_k(p_B) \right|^2, \quad c = (1, 1), \quad (18)$$

with  $\alpha_k$  and  $\beta_k$  the part  $k$ 's amplitude at the port on each side and  $e_k(p)$  the port's integer coefficient on the part  $k$ , the window's sum outside the square as the implementation's gate computes it, on both members of the level pair. The joint share is never the square of a local sum, which one detector could form alone and which gives  $S$  at most 2. One draw is made per pair, the detectors' draw with the declared seed, the GameBoard drawing nothing: the first side from its marginal and the partner from the conditional; each side's write at the Node of its exchange spreads by Rule3 and changes no number of the row, one click per record (Section 5.4) (ASSUMPTION, CLICKS) [3, Bell in nature's own layout].

### 9.4 Bell: what the model says it must give

(1) Rule3 gives each part's amplitudes at the ports: the two parts of one beam, laid equal, are two records of one family stepped by the same line along the same path, identical bit for bit (THEOREM, GAMEBOARD). (2) The meeting with the linearity of Rule3 gives the joint amplitude: the realised click at A's port weights the parts by A's analyser, is read back to the root where the two parts were one event and forward to B, where B's analyser weights them, and the root sums the two branches,  $\sum_k c_k e_k(a) e_k(b) = \cos(a - b)$  with  $c$  the parts' laid weights,  $(1, 1)$  for the symmetric lay (DERIVED, CLICKS). (3) That the share is the density of the meeting, its square, is Born's rule for a pair [44]: nothing in Rule3 forces it; everything after it, the fractions,  $S$ , GHZ's  $M$ , is derived in the algebra (ASSUMPTION, FITTED; CLICKS). From the three, with the pair's common phase  $\theta$  uniform over the pairs and A's + port at  $a$  and B's at  $b$ ,

$$\begin{aligned} \text{the meeting:} & \quad P(++) = \frac{1}{2} \cos^2(a - b), \quad E(a, b) = \cos 2(a - b), \quad S = 2\sqrt{2}; \\ \text{a local credit by the shares:} & \quad P(++) = \frac{1}{4} + \frac{1}{8} \cos 2(a - b), \quad E = \frac{1}{2} \cos 2(a - b), \quad S = \sqrt{2}; \\ \text{a local credit by the sign:} & \quad E = 1 - 4|a - b|/\pi \quad (|a - b| \leq \pi/2), \quad S = 2; \\ \text{the contrast credit, one side weighted:} & \quad E = \cos 2(a - b) \text{ at one side's efficiency } 2/\pi, \quad S = 2\sqrt{2}, \end{aligned} \quad (19)$$

The second, third and fourth lines follow by averaging over the pair's phase  $\theta$  (DERIVED, CLICKS; S.10); the fourth counts one side weighted by its contrast and the other side's sign always, by  $\langle \cos x \operatorname{sgn} \cos(x - 2\delta) \rangle = (2/\pi) \cos 2\delta$ , the symmetric readings giving no cosine (S.10). So  $E(a, b) = \cos 2(a - b)$  under the meeting. For the declared pairs it is exact, by Lagrange's identity of the reader on equal parts (S.10), so at the four settings  $E = 119/169$ ,  $-119/169$ ,  $120/169$  and  $120/169$  and  $S = 478/169 = 2.8284$  exactly, an efficiency of 1 by construction and the marginal one half on each side whatever the partner's setting (COMPUTED, CLICKS; S.10). For unequal parts,  $r$  their products' ratio at the ports, the same credit gives  $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$  with  $\rho = 2r/(1 + r^2)$  and  $S = \sqrt{2}(1 + \rho)$  at the ideal angles (DERIVED; CLICKS). The maximum of  $S$  over the four settings is  $2\sqrt{1 + \rho^2}$ , Tsirelson's bound [77] at  $\rho = 1$  and never the algebraic 4; the declared pairs' 478/169 sits below  $2\sqrt{2}$  by the rational angles (DERIVED, CLICKS).

What the GameBoard gives without the declaration is the second and third lines of Eq. (19), two results on local credits on the same world (DERIVED under the uniform phase, CLICKS): a credit of each side by the shares of its ports gives  $S = \sqrt{2}$ , and a credit by the sign of its larger port gives the sawtooth, whose sum is exactly 2, Bell's bound. Bell's value therefore sits in the credit rule, a declaration, and is no independent evidence that the rule accounts for entanglement. The same credit on four lines gives the three-quantum GHZ correlation [76]  $E_3 = \cos 2(a + b + c)$ ,

every pairwise  $E$  0 and Mermin's  $M = -4$  where every local credit gives at most 2 (DERIVED under the declared credit, CLICKS; S.40) [1, the GHZ gate]. No other entanglement than the pair's and the swap's is named as met (S.51). Between separately prepared bodies joined by a coupling or by a transferred quantum the law's credits give at most 2, where nature's gate-joined ions [32] and transmons with a transferred photon [34] exceed it: a finding of the first rank against the law as written (S.56 (14)). No signal passes between the sides, each side's marginal being one half; no settings are chosen by the GameBoard's own past, so a superdeterministic world [65] is not this one [1, the rows against nature, (h)][3, the click is the meeting of the future with the past; Bell in nature's own layout].

## 10 Discussion: the constants, the calibration and the findings

### 10.1 The constants

The quantum of action is one click, whose energy is the quantum's norm  $T$ . Newton's constant is not declared: a body whose wells are  $s$  quanta per interval makes the massless field's rest  $3G(r)s$  at  $r$  Links over the field's divisor,  $3s/(4\pi r)$  far away, and the level slows the clock by the potential  $\Phi = -c/\Gamma = -U$ , Newton's sign, so (DERIVED, GAMEBOARD; S.25)

$$\boxed{\frac{G_{\text{clock}}}{c^2} = \frac{3}{4\pi \Gamma E_g}, \text{ that is } G_{\text{clock}} = \frac{1}{4\pi \Gamma E_g} \text{ per quantum of well, in Links and intervals}}, \quad (20)$$

a reading of  $\Gamma$ , the file's  $E_g$  and the lattice. Kepler's  $G$ , read from the orbits of matter as clicks, is  $[2 \cos \omega_0 / (1 + \cos \omega_0)] G_{\text{clock}}$  by Eq. (14), 0.8 of it at  $[2, 3]$  (DERIVED, CLICKS). The vacuum content is 0 in the file, a declaration no click bounds (Section 6.5), and a cosmological constant, if the law has one, would be the vacuum write's rate, a hypothesis not claimed here (S.29), not this rest [1, the constants].

**The fine-structure constant, a coefficient of the file; the units.** Under the holder's rotation (Section 5.5) the quantum of a family is one unit of the invariant,  $2A^2 \sin \omega_s = T$ , its energy  $T \sin \omega_s$  and its Wronskian  $A^2 \sin \omega_s = T/2$  for every family, the unit of charge, so charge universality is derived from the writes,  $|q_p| = |q_e|$  with nothing declared, nature's to  $10^{-21}$  (DERIVED; CLICKS; S.26). The holder of the sign carries one declared integer weight  $k$  in its write, the written level  $k$  times the plain one and the turn the level over  $\Gamma$ , as gravity's row carries  $E_g$ , and the law's coupling between two quanta is  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi)k/(\Gamma E_s)$ , independent of the family's gap (DERIVED under the rotation,  $k$  ASSUMPTION; CLICKS; S.26). The force on a body of  $n$  quanta and sense  $\pm$  is  $\pm n \nabla \theta$ , Coulomb's  $q_1 q_2$  with  $q = \pm ne$ ,  $e$  one quantum's Wronskian  $T/2$ ; the reader sources the holder it reads, so the pair is reciprocal (DERIVED; S.26, S.43). Nature's  $\alpha = 1/137.036$  fixes  $k/(\Gamma E_s) = 1/28.33$  and nothing else:  $\alpha$  is a coefficient of the file, read from one click as  $G$  is, neither derived nor a finding. A charged body and its light conserve by a line of the law,  $E_s T \cos \omega_s = k_w \Gamma$ , the body's loss under the turn and the holder's gain under the write agreeing if and only if it holds (DERIVED for the identity under the implementation's present convention of the turn; the line ASSUMPTION; S.26); under it  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi)T \cos \omega_s / \Gamma^2$ , at nature's gap  $T = \Gamma^2/28.33$ ,  $E_s$  a unit of the level and the file one integer fewer. The line binds the plane families that read the holder of the sign and not light or a real-line family; the rule's own universe has no plane family, so its  $T = 2^{15}$  stands, and the implementation's charged worlds were laid before the line and do not satisfy it, their checks reading the static force and the sense alone (one derivation; S.26). The two held rows are one kind of row with one kernel, the sign holder reading a quantum's Wronskian  $T/2$  and the content holder its form  $\sin \omega_s$ , so gravity's coupling between two quanta of the family  $s$  is  $\alpha_g(s) = (3\sqrt{3}/4\pi) \sin^2 \omega_s / (\Gamma E_g)$ , and the ratio of the two forces at that family is  $F_e/F_g = k E_g / (2E_s \sin^2 \omega_s)$ ; between two families the weights cancel,  $(F_e/F_g)_e / (F_e/F_g)_p = (m_p/m_e)^2$  with no number, nature's  $4.17 \times 10^{42} / 1.24 \times 10^{36} = 1836.15^2$ , an identity forced by charge universality and  $F_g \propto m^2$  and no test (DERIVED; CLICKS; S.44). On the board there is no  $10^{42}$ : the number appears when a detector compares the two forces on one quantum, most of it the smallness of the electron's energy against its unit of charge (S.44). The units are the electron's: one interval is  $\omega_e \hbar / (m_e c^2) = 1.288 \times 10^{-21} \text{ s} \times \omega_e$  and one Link  $\sqrt{3} \omega_e \bar{\lambda}_e = 6.69 \times 10^{-13} \text{ m} \times \omega_e$ . What a click reads of a quantum of frequency  $\omega$  is its share,  $T \sin \omega$  (S.14), at most  $T$ , against the electron's rest energy  $T \sin \omega_e$ ; LHAASO's photons of 1.4 PeV [66], read by a calorimeter as a sum of such shares, give  $\omega_e < 3.7 \times 10^{-10}$ ,  $\text{den} > 1.5 \times 10^{19}$  and  $\gamma_K - 1 = \text{den}/\text{num} - 1 = \omega_e^2/2$  below  $6.7 \times 10^{-20}$  for every orientation of the lattice (the band's top,  $\omega = \pi$  along the diagonal, is a GameBoard statement and no click's energy) [1, what a click reads of a quantum], and the Link's bound of Section 10.2 for a beam along an axis, the Link over the unit,  $\omega_e < 2.2 \times 10^{-14}$  and  $\gamma_K - 1 < 2.4 \times 10^{-28}$  (DERIVED, two derivations; CLICKS; S.24, S.27).  $\Gamma$  reads nothing of  $\omega_e$  or of  $\alpha$  and is bounded by nothing of the clocks, which read the mean of the write's dithered sub-unit level, the exact rest, and not  $\Gamma$  (an argument on the mean, the dynamics under a dithered level open; S.26);  $\Gamma$  is bounded by no click of the clocks; the gap bound binds the matter family's own denominator,  $\text{den} > 1.5 \times 10^{19}$  for the least pair, and  $\Gamma$  only in a file that writes that pair over  $\Gamma$ , as the shipped file does; the energy line ties it to  $T$ ,

**Table 4** The calibration series: each coefficient of the universe’s file, the one click of nature that reads it, and the layer it lives in (DERIVED for every form of reading, the integer the file’s). The lattice reads none of them.

| The coefficient                         | The one click that reads it                                                                                                          | The layer                                |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| $T$ , the unit of action                | Planck’s constant: the counter’s unit of credit and Millikan’s slope (Section 7.4)                                                   | the click                                |
| The Link and the interval               | $\omega_e$ ’s bounds from the highest photon, LHAASO’s 1.4 PeV, and light’s dispersion, MAGIC’s bound [67] (S.24, S.27)              | the board, through the electron’s family |
| $\Gamma$ , the board’s clock            | bounded by no click of the clocks (Section 10.1); the energy line ties it to $T$ , $E_s$ and $k_w$ (S.26)                            | the board                                |
| The gaps, one per family                | the masses: $\omega_e$ by the electron’s mass, $g_p/g_e = (m_p/m_e)^2$ under one denominator, read as click masses $T \sin \omega_s$ | the board, through bound bodies          |
| $E_g$ , gravity’s weight                | Newton’s $G$ , the fall and the orbits, Eq. (20)                                                                                     | the board, through the fall              |
| $k$ , the sign holder’s weight          | Coulomb’s force between two quanta, $k/(\Gamma E_s) = 1/28.33$ (S.26)                                                                | the board, through two free quanta       |
| $c_{\text{vac}}$ , the vacuum content   | declared 0; no click reads it, every row reading the content; GW170817 bounds $\gamma_{\text{GW}} - \gamma_{\text{EM}}$              | the board                                |
| The nuclear holders’ ranges and weights | the nuclear binding and the nuclei’s sizes, if the holders are admitted (Section 10.3; HYPOTHESIS)                                   | the board, through the nucleus           |

$E_s$  and  $k_w$  (CLICKS) [1, a body and its light keep one energy; the law’s alpha is a coefficient of the file; the units are the electron’s and the band’s top is the highest photon energy; what is open, item 19][3, the strong force’s kernel derived: the colour coupling at the Link is the charge’s alpha, no running, not the strong force].

## 10.2 Nature’s numbers enter only as clicks

**What the lattice fixes with no number.** Before any file the lattice fixes the geometry with its group of 48, hence one preferred frame and  $\alpha_1 = -8$ ; the dynamics, reversible to the bit with no loops, hence no running of  $\alpha$  and no Lamb shift; the conservations per family; the band, light at  $1/\sqrt{3}$  Link per interval under the causal bound 1; the paces, the exponential metric with the finest level  $1/\Gamma$ ; the holders’ kernel  $3/(4\pi r)$ ; the click layer’s bilinear credit; and one scale  $\Gamma$  with pure numbers of order one, every large number a power of  $\Gamma$  or a declaration, each with its status in S.54. In one sentence: the lattice is a machine of forms; it fixes every form and every ratio of order one, conserves, reverses and makes no noise; it fixes no strength, no scale and no large number; those are the file’s [1, the properties of the lattice].

**The calibration series.** Every measured number of an experiment is first written as a click pattern on a declared detector against a declared clock; the families’ pairs and the units  $\Gamma$  and  $T$  are fixed by that match and by nothing read off the GameBoard, the two ratios the model fixes without it being light’s speed of  $1/\sqrt{3}$  Links per interval and the causal bound of 1 (ASSUMPTION, CLICKS) [1, the reading; nature’s numbers enter the board only as clicks]. No coefficient of the file is computed from the lattice: each is read by one click of nature on a bound body, the lattice fixing the form of the reading and the file holding the integer, Table 4 (S.54); no number comes from the lattice itself (HYPOTHESIS for any such number, S.54) [1, the calibration series]. One check beside the table, the two clocks: a cavity of bodies and an atom’s beat shift alike by  $N$  at long wavelength, as nature’s clocks do, atom against atom to  $10^{-6}$  and cavity against maser to  $10^{-4}$  [38], a cavity of declared faces at  $N^2$  (DERIVED; CLICKS; S.31).

**One prediction, untested, one check and one open question.** One number follows from the law’s lines where general relativity and quantum theory do not meet, entered as an explicitly named hypothesis, read by no click yet (DERIVED, HYPOTHESIS; S.41): *the gravity of light* (CLICKS, untested), a light packet pulling at  $1 \times E/c^2$ , where general relativity’s pencil pulls at  $2 \times E/c^2$  [69]. The check: a bound body laid as one record on two paths at two heights has its beat scale as  $N = 1 - U$ , so the fringes’ visibility falls as  $|\cos(\Omega \Delta U T/2)|$ , 0.989 for a strontium clock at one metre for one second, the semiclassical expectation’s magnitude [68], unmeasured (CLICKS; S.41). The open question: a collapse ends in frozen clocks that read nothing, are read, and are dark by the share’s declaration (Section 7.2); whether such a region radiates, Hawking’s temperature [73] and its entropy are open, the argument for  $T_H = 0$  withdrawn after the stencil’s reading (S.41, Section 10.3).

## 10.3 Findings against nature and open questions

**GameBoard readings, not findings.** The rule’s own universe at [2, 3] is no finding against nature, which measures at its own pair (GAMEBOARD; Section 8, (e)). The masses are families: nature’s mass ratios are not whole numbers ( $m_p/m_e = 1836.153$ ), so no particle of nature is a cluster of the electron’s quanta, and the proton is one quantum of its own family, read by hydrogen’s clicks as one unit of  $W$  over 0.84 fm (DERIVED for the ratios, ASSUMPTION for the

families; CLICKS; S.53); charge universality and the equivalence between families follow from the writes (DERIVED); if the families share one denominator, the masses are the square roots of the gaps (HYPOTHESIS). The model's particle physics is families with declared masses and conversions by click [3, the proton's two readings].

**Findings against nature.** The massless field's travelling events carry the breathing polarisation and the lattice-aligned + and no  $\times$ , where nature's gravitational waves carry + and  $\times$  at every orientation (CLICKS) [1, what is open, item 23]. Gravity's readers are real records with no current's read, so a gyroscope precesses by 0 from a rotating body's motion, where Gravity Probe B measured  $37.2 \pm 7.2$  milliarcseconds per year against general relativity's 39.2 [74] (DERIVED, CLICKS) [3, general relativity in the law; the current's read]. A metric with no  $g_{0i}$  in its preferred frame has  $4\gamma + 4 + \alpha_1 = 0$ , so  $\alpha_1 = -8$  at  $\gamma = 1$  against nature's  $|\alpha_1| < 4 \times 10^{-5}$  from the binary pulsar PSR J1738+0333 [75], a finding (DERIVED, CLICKS; S.35) [1, what is open, item 15]. With  $g_{0j} = 0$  in the lattice frame,  $g_{00} = -e^{-2U}$  and  $g_{ij}$  isotropic, the standard post-Newtonian gauge, the two  $g_{0j}$  coefficients vanish separately and give  $\alpha_1 = -8$  and  $\alpha_2 = -1$  together, their sum the gauge-invariant statement above, against  $|\alpha_2| < 1.6 \times 10^{-9}$  from the Sun's spin [60], conditional on the metric read for a static source, like  $\alpha_1$  (named, not claimed). At the second order in  $z$  the ball of dust alone decelerates,  $q_0 = \Omega_m/2 > 0$ , where the supernovae read  $q_0 \approx -0.5$ , flat  $\Lambda$ CDM's  $\Omega_m/2 - \Omega_\Lambda$  [53][54][55]; the paper claims nothing beyond  $z \ll 1$  there, and the law's documents carry by name the hypothesis that would close it [1, the vacuum writes], which is not this paper's claim (CLICKS).

**The lattice's frame, and its mend as a hypothesis.** At long wavelength every record's band is a Lorentz form,  $\omega^2 = \omega_0^2 N^2 + c_m^2 k^2$ , so the moving clock's  $\gamma$  follows and no Michelson-Morley shift appears at nature's pair, to  $10^{-19}$  (at the computed pair  $c$  and  $c_m$  differ, and a matter-armed interferometer would shift at order  $v^2$ ), while the content's Poisson rest in the lattice's frame gives a moving mass no  $g_{0a}$  and no  $g_{ab}$  (DERIVED; CLICKS) [1, the lattice's frame is read by gravity alone, and its carrier is a pair of Links; what is open, item 15]. The mend, an explicitly named hypothesis with no line written (HYPOTHESIS): a carrier on a pair of Links sourced by the mass current; against it, no real, local, explicit, bijective lattice line on Rule3's clock carries  $g_{0a}$  among the five forms tried, the four on the synchronous clock closing as one and the fifth, the staggered clock, failing the redshift (DERIVED for the five forms, no theorem over every form; S.36). [3, nothing that is not Rule3 remains]

**The strong force, by derivation.** The strong force is two things: confinement, the click's indivisibility of a record, derived under a family of dimension 3, an explicitly named hypothesis (Section 3.2), with no coupling; and the nuclear binding, a declared pair of Yukawa holders read by the nucleon families alone, under which the nucleus's size, binding and its light are derived in form and its shells are not. Confinement is not a force: a click credits a whole record and never one line of it (Section 5.3), so nothing on the board is a quark to free, and a hard click reads three fractions summing to one, the lines' shares, unchanged over the momentum transfer: Bjorken scaling exact, where nature reads its logarithmic violations, an explicit finding (DERIVED under the dimension-3 hypothesis; CLICKS; S.37, S.56). The nuclear binding is reached by no derived body of the law (CLICKS): a body's binding per quantum is bounded by  $0.76 CW/\Gamma$  at one Node and is about  $0.36 CW/(\Gamma r)$  for a uniform ball of  $r$  Links (DERIVED, a bound; GAMEBOARD; S.38), and under the families' reading the nucleon is one quantum whose binding is the nuclear holder's well,  $W \geq 2.35 \Gamma \kappa / \omega_p^3$  (S.45). The nucleus is therefore a hypothesis under its own name: a bound body of nucleon quanta in the well of a nuclear holder, Yukawa's kernel (S.39) with its range and weight declared, read by the nucleon families alone (Section 7), and a core holder of shorter range and negative weight, four declared integers per nucleus (HYPOTHESIS). Under it the threshold, the binding at a body's size, fusion and fission with the shape of Bethe and Peierls [46], and the size law are computed in form (one derivation; S.45). The findings: one well gives  $R \propto 1/A$  where nature's nuclei grow as  $A^{1/3}$  and saturate, the core holder the saturating declaration; the shells and the magic numbers (Pauli's exclusion) and the neutron's lifetime are explicit findings (CLICKS). [1, the strong force; the nuclear holder; the nucleus; what is open, item 1][3, is the strong force not events that do not split].

**The weak force.** The model has no electroweak theory; its weak interaction is the conversion, a click, an explicitly named hypothesis [1, the conversion]. Every rotation holder, massive or not, couples at  $\alpha$  at the Link, its gap its range and never its strength (DERIVED; GAMEBOARD; S.39); no  $G_F$  follows. A three-line record read as  $(\nu, e)$  under one  $O(3)$  has no fourth rotation, so no  $Z$ , where nature has the neutral current [78] and the  $Z$  [79]; parity is conserved, where nature's weak interaction violates it [80]; and the law has no spontaneous click, so the conversion has no rate and the neutron's 880 s no source: three explicit findings (CLICKS). What decays in the model is a body and never a quantum (Section 7), so the neutron's end is the conversion [3, the weak-force reading].

**The gaps in two columns.** The first column is the world's, and the method invites another world:  $\alpha$  through  $k$ ,  $G$  through  $E_g$ , the Link and the masses, the hierarchy of the two weights, the spin holder and the click's unit. The second is the rule's, each with its root, none a number's fault [3, why the gaps]: the nuclear binding by any derived body (one denominator reads every level into the paces; DERIVED, a bound); frame dragging,  $\alpha_1$  and the

$\times$  polarisation (no line carries  $g_{0a}$  among the five forms tried; DERIVED);  $\alpha$ 's running [79], the Lamb shift and the lifetimes (no loops, no spontaneous click);  $g - 2$  and Pauli's statistics (no loops; no vector gives the exclusion); the  $Z$  and parity (the step commutes with the cube's reflections).

**The record's parts are its particles.** A hypothesis by name beside the families' reading (S.56 (7)), which the engine's gate decides (HYPOTHESIS; CLICKS). Nature's quanta are the parts of the matter records and the records of the three holders: the baryons the parts of the three-line record (S.37); the charged leptons families of the plane record and the neutrino a real-line family of matter, neutral by the law's own line, the pair converted into each other only at the conversion click; light, the binding holder's quanta and the nuclear holders' quanta the holders' own records. A quantum changes part or family only at a click, the conversion, by a declared table of which part gives which, at a declared rate, the rates read from nature as the lifetimes are until a derivation; the neutrino meets no transition of nature's detectors, bodies of charged matter, except through the conversion, and a declared detector of its family would click it by the law's one click; postulate 2 holds, a Node carrying its parts' numbers and nothing else. Its test is the neutron's end, the conversion of one part at the declared rate (880 s), the gate world of the giving click; it gives neither the rates nor the mixing nor parity's violation, and this section's findings stand [1, the hypotheses under their own names].

**Open.** The hypothesis that only a click moves a quantum between families (Section 5.4; HYPOTHESIS), with its gate worlds below. No standing body is computed in this round, the three problems of its computation and the fix named (Section 7.3); the black hole's quanta, whether a region of frozen clocks radiates, with Hawking's temperature and the entropy beside it (S.41). The shelved ion's telegraph [70], bimodal counts with exponential dark periods, which the law without the instruments' writes fails (Poisson at half the rate) and the body's own giving click as the jump operator meets, the body re-laid with its whole count in the lower mode at a declared rate, the shelving the body's taking click of the weak drive's quantum at the Zeno-limited rate  $\Omega_w^2/R$  [72], the bright periods' mean  $R/\Omega_w^2$  and the dark the declared  $\tau_D$ , the statistics' shape derived under the declared rate from the rate equations' known form [71] and the Zeno-limited rate's known form [72] (DERIVED under a declaration), the rate itself declared and not derived (HYPOTHESIS under its own name; S.56 (15)); the three gate worlds of the click's write, the ion's body with its modes as parts and two declared rates, the anticoincidence's (S.57) and the Zeno world's (S.59), are derived here as the blind rows of their worlds, the runs the gate after the engine's freeze, and until it nothing of them is claimed; a gate-built register stands outside by postulate 2 (Section 3.6). Three routes stand to be checked in the implementation, explicitly named hypotheses (HYPOTHESIS; GAMEBOARD until run): the nucleus as a bound body in the declared nuclear holders, its form factor against a one-event record's flat  $F = 1$ , nothing of it run; the weak interaction as the conversion at a click,  $G_F$  the click. An exactly conserved integer charge; gravity's divisor  $E_g$  (EXPERIMENT); the matter pair's gap; what fixes  $\Gamma$ ; a current's read for real records [1, what is open]. The relations among the file's integers are the next paper's program, each an explicitly named hypothesis tested by clicks: one denominator for all the families, the masses the square roots of the gaps; whether  $k$ ,  $E_g$ ,  $T$  and  $\Gamma$  are tied, the energy line of Section 10.1 the first of them and already a law (HYPOTHESIS) [1, no free number but the mass]. One finding named and not claimed: the law's resonance adds frequencies,  $\omega_L = \omega_j - \omega_m$  (S.43), and its share adds sines,  $T(\sin \omega_j - \sin \omega_i)$  (S.14); the two agree to  $\omega^2/6$  relative, below  $10^{-18}$  at nature's gap, and part at the Link scale, where which of the two a click takes the law as written does not decide; no click of nature reaches there [1, what is open, item 39]. The atom. The law does not support the reduced mass: a Node holds no joint amplitude of two records (postulate 2 of Section 3.6), so a product of records about one centre has no relative motion of the pair; under the second of the two lines decided for the law (S.60) the nucleon's centre is one classical variable; the isotope shifts and the muonic atoms are therefore outside what the law derives. The many-body correlation is not reproduced by a product of records (the ground levels, a GameBoard reading: helium 1.4 percent, positronium 56 percent; the clicks in S.60's table), and Pauli's exclusion has no line (S.46, S.60; the engine round open) [1, the atom is a bound body of the holder of the sign].

## 11 Conclusion

One line of integer arithmetic at every Node of a cubic lattice is reversible to the bit and conserves a quadratic form with its current. A click is the meeting of the events run forward from an emitter, the future, with the same line read backward from the click, the past, at a declared detector, the choice among the meetings the one non-local element. Under the one assumption the weak-field forms are Einstein's in the clock's potential, the shadow's capture radius a prediction, and against Kepler's potential the matter band's factors reach 1 only as the family's gap closes. The paper claims the method and not nature; it is offered for the reader's judgement, test and further use, one world of the many the method admits. Between electricity and gravity the number is in the click, not on the board. What stands after the checks is the core, exact and reversible; the weak field from the composed paces under one named assumption;

the click with its one non-local place; the marks; and the recomputability of every number from the public files, all derived from one chosen set of rules of the many the engine admits. Why nature is so, we do not claim to know.

## Statements and Declarations

**Funding.** None. **Competing interests.** None. **Ethics approval and consent.** Not applicable. **Data and code availability.** The law's document, the implementation and the figures' scripts are developed in the public Git repository <https://github.com/Closer24/Universe24> and archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [43], MIT licence. **Author contributions.** The author laid down every postulate and decision of the model, directed the work and wrote the paper. **The use of a large language model.** The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author's direction, for the software, the computations, the derivations and their checks, and the drafting of the text; the model is not an author, and the author reviewed every statement and takes full responsibility for the content.

## References

- [1] The law of the GameBoard, `docs/ALGEBRA.md` of the archive [43], cited by the name of its section: the GameBoard and the families (the objects, a family's declaration, the interval, readings and measurements); Rule3 (the line, the direction, the four acts, the paces, the bound, the self-source, the conserved form); the click (the booking, the count is the record's share, the click is the detector's report, no click names a Node, the algebra of the exclusivity); the primitives; the stable body (what a body is, the well, the functional and the dilation, the surplus leaves, the generator, the velocity); what the law says of nature (the reading, with no waves on the event board, what produces a click is not known and nature's numbers enter the board only as clicks; the hypotheses under their own names; the postulates; the constants; the rows against nature); what is open.
- [2] The engine, `docs/ENGINE.md` of the archive [43]: the words, the NodeState, the interval, the loader and the files, the output, how to run a world, how to add a family, the gates.
- [3] The decisions, `docs/HIGHLIGHTS.md` of the archive [43]: the model's decisions in force, one line each, cited by the line's name.
- [4] K. Zuse, *Rechnender Raum* (Vieweg, Braunschweig, 1969); English translation: *Calculating Space*, MIT Technical Translation AZT-70-164-GEMIT (MIT Project MAC, Cambridge, 1970).
- [5] E. Fredkin and T. Toffoli, Conservative logic, *International Journal of Theoretical Physics* 21, 219 (1982), doi:10.1007/BF01857727.
- [6] N. Margolus, Physics-like models of computation, *Physica D* 10, 81 (1984), doi:10.1016/0167-2789(84)90252-5.
- [7] T. Toffoli and N. Margolus, Invertible cellular automata: a review, *Physica D* 45, 229 (1990), doi:10.1016/0167-2789(90)90185-R.
- [8] S. Wolfram, *A New Kind of Science* (Wolfram Media, Champaign, 2002).
- [9] G. 't Hooft, Duality between a deterministic cellular automaton and a bosonic quantum field theory in 1+1 dimensions, *Foundations of Physics* 43, 597 (2013), doi:10.1007/s10701-013-9702-z.
- [10] G. 't Hooft, The Cellular Automaton Interpretation of Quantum Mechanics, *Fundamental Theories of Physics* 185 (Springer, Cham, 2016), doi:10.1007/978-3-319-41285-6.
- [11] I. Bialynicki-Birula, Weyl, Dirac, and Maxwell equations on a lattice as unitary cellular automata, *Physical Review D* 49, 6920 (1994), doi:10.1103/PhysRevD.49.6920.
- [12] D. A. Meyer, From quantum cellular automata to quantum lattice gases, *Journal of Statistical Physics* 85, 551 (1996), doi:10.1007/BF02199356.
- [13] P. Arrighi, An overview of quantum cellular automata, *Natural Computing* 18, 885 (2019), doi:10.1007/s11047-019-09762-6.
- [14] T. Farrelly, A review of quantum cellular automata, *Quantum* 4, 368 (2020), doi:10.22331/q-2020-11-30-368.
- [15] G. M. D'Ariano and P. Perinotti, Derivation of the Dirac equation from principles of information processing, *Physical Review A* 90, 062106 (2014), doi:10.1103/PhysRevA.90.062106.
- [16] R. P. Feynman, Simulating physics with computers, *International Journal of Theoretical Physics* 21, 467 (1982), doi:10.1007/BF02650179.
- [17] J. A. Wheeler and R. P. Feynman, Interaction with the absorber as the mechanism of radiation, *Reviews of Modern Physics* 17, 157 (1945), doi:10.1103/RevModPhys.17.157.
- [18] J. G. Cramer, The transactional interpretation of quantum mechanics, *Reviews of Modern Physics* 58, 647 (1986), doi:10.1103/RevModPhys.58.647.
- [19] Y. Aharonov, P. G. Bergmann and J. L. Lebowitz, Time symmetry in the quantum process of measurement, *Physical Review* 134, B1410 (1964), doi:10.1103/PhysRev.134.B1410.
- [20] Y. Aharonov and L. Vaidman, The two-state vector formalism: an updated review, *Lecture Notes in Physics* 734, 399 (Springer, Berlin, 2008), doi:10.1007/978-3-540-73473-4\_13.
- [21] H. Price, *Time's Arrow and Archimedes' Point: New Directions for the Physics of Time* (Oxford University Press, New York, 1996).
- [22] K. B. Wharton and N. Argaman, Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics, *Reviews of Modern Physics* 92, 021002 (2020), doi:10.1103/RevModPhys.92.021002.
- [23] N. Harrigan and R. W. Spekkens, Einstein, incompleteness, and the epistemic view of quantum states, *Foundations of Physics* 40, 125 (2010), doi:10.1007/s10701-009-9347-0.
- [24] J. S. Bell, On the Einstein Podolsky Rosen paradox, *Physics Physique Fizika* 1, 195 (1964), doi:10.1103/PhysicsPhysiqueFizika.1.195.
- [25] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, Proposed experiment to test local hidden-variable theories, *Physical Review Letters* 23, 880 (1969), doi:10.1103/PhysRevLett.23.880.
- [26] A. Fine, Hidden variables, joint probability, and the Bell inequalities, *Physical Review Letters* 48, 291 (1982), doi:10.1103/PhysRevLett.48.291.
- [27] P. M. Pearle, Hidden-variable example based upon data rejection, *Physical Review D* 2, 1418 (1970), doi:10.1103/PhysRevD.2.1418.
- [28] P. H. Eberhard, Background level and counter efficiencies required for a loophole-free Einstein-Podolsky-Rosen experiment, *Physical Review A* 47, R747 (1993), doi:10.1103/PhysRevA.47.R747.
- [29] A. Garg and N. D. Mermin, Detector inefficiencies in the Einstein-Podolsky-Rosen experiment, *Physical Review D* 35, 3831 (1987), doi:10.1103/PhysRevD.35.3831.
- [30] B. Hensen et al., Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres, *Nature* 526, 682 (2015), doi:10.1038/nature15759.
- [31] M. Giustina et al., Significant-loophole-free test of Bell's theorem with entangled photons, *Physical Review Letters* 115, 250401 (2015), doi:10.1103/PhysRevLett.115.250401.
- [32] M. A. Rowe, D. Kielpinski, V. Meyer, C. A. Sackett, W. M. Itano, C. Monroe and D. J. Wineland, Experimental violation of a Bell's inequality with efficient detection, *Nature* 409, 791 (2001), doi:10.1038/35057215.
- [33] L. K. Shalm et al., Strong loophole-free test of local realism, *Physical Review Letters* 115, 250402 (2015), doi:10.1103/PhysRevLett.115.250402.
- [34] S. Storz et al., Loophole-free Bell inequality violation with superconducting circuits, *Nature* 617, 265 (2023), doi:10.1038/s41586-023-05885-0.
- [35] T. Regge, General relativity without coordinates, *Il Nuovo Cimento* 19, 558 (1961), doi:10.1007/BF02733251.
- [36] L. Bombelli, J. Lee, D. Meyer and R. D. Sorkin, Space-time as a causal set, *Physical Review Letters* 59, 521 (1987), doi:10.1103/PhysRevLett.59.521.
- [37] P. Touboul et al. (MICROSCOPE Collaboration), MICROSCOPE mission: final results of the test of the equivalence principle, *Physical Review Letters* 129, 121102 (2022), doi:10.1103/PhysRevLett.129.121102.
- [38] C. M. Will, The confrontation between general relativity and experiment, *Living Reviews in Relativity* 17, 4 (2014), doi:10.12942/lrr-2014-4.

- [39] J. Collins, A. Perez, D. Sudarsky, L. Urrutia and H. Vucetich, Lorentz invariance and quantum gravity: an additional fine-tuning problem?, *Physical Review Letters* 93, 191301 (2004), doi:10.1103/PhysRevLett.93.191301.
- [40] D. Mattingly, Modern tests of Lorentz invariance, *Living Reviews in Relativity* 8, 5 (2005), doi:10.12942/lrr-2005-5.
- [41] R. Courant, K. Friedrichs and H. Lewy, Über die partiellen Differenzengleichungen der mathematischen Physik, *Mathematische Annalen* 100, 32 (1928), doi:10.1007/BF01448839.
- [42] H. Yilmaz, New approach to general relativity, *Physical Review* 111, 1417 (1958), doi:10.1103/PhysRev.111.1417.
- [43] A. Gonen, Universe24: a discrete simulator of local physical laws (Reality Theory), Zenodo (2026), doi:10.5281/zenodo.22738746 (the concept DOI, resolving to the latest version).
- [44] M. Born, Zur Quantenmechanik der Stoßvorgänge, *Zeitschrift für Physik* 37, 863 (1926), doi:10.1007/BF01397477.
- [45] P. Grangier, G. Roger and A. Aspect, Experimental evidence for a photon anticorrelation effect on a beam splitter: a new light on single-photon interferences, *Europhysics Letters* 1, 173 (1986), doi:10.1209/0295-5075/1/4/004.
- [46] H. A. Bethe and R. Peierls, Quantum theory of the dipole, *Proceedings of the Royal Society A* 148, 146 (1935), doi:10.1098/rspa.1935.0010.
- [47] H. Yukawa, On the interaction of elementary particles. I, *Proceedings of the Physico-Mathematical Society of Japan* 17, 48 (1935), doi:10.11429/ppmsj1919.17.0-48.
- [48] B. P. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, Fermi Gamma-ray Burst Monitor and INTEGRAL), Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A, *The Astrophysical Journal Letters* 848, L13 (2017), doi:10.3847/2041-8213/aa920c.
- [49] G. H. Derrick, Comments on nonlinear wave equations as models for elementary particles, *Journal of Mathematical Physics* 5, 1252 (1964), doi:10.1063/1.1704233.
- [50] S. I. Pohozaev, Eigenfunctions of the equation  $\Delta u + \lambda f(u) = 0$ , *Soviet Mathematics Doklady* 6, 1408 (1965).
- [51] S. I. Pekar, Untersuchungen über die Elektronentheorie der Kristalle (Akademie-Verlag, Berlin, 1954).
- [52] W. H. McCrea and E. A. Milne, Newtonian universes and the curvature of space, *The Quarterly Journal of Mathematics* os-5, 73 (1934), doi:10.1093/qmath/os-5.1.73.
- [53] A. G. Riess et al., Observational evidence from supernovae for an accelerating universe and a cosmological constant, *The Astronomical Journal* 116, 1009 (1998), doi:10.1086/300499.
- [54] S. Perlmutter et al., Measurements of Omega and Lambda from 42 high-redshift supernovae, *The Astrophysical Journal* 517, 565 (1999), doi:10.1086/307221.
- [55] D. Brout et al., The Pantheon+ analysis: cosmological constraints, *The Astrophysical Journal* 938, 110 (2022), doi:10.3847/1538-4357/ac8e04.
- [56] R. V. Pound and G. A. Rebka, Apparent weight of photons, *Physical Review Letters* 4, 337 (1960), doi:10.1103/PhysRevLett.4.337.
- [57] B. Bertotti, L. Iess and P. Tortora, A test of general relativity using radio links with the Cassini spacecraft, *Nature* 425, 374 (2003), doi:10.1038/nature01997.
- [58] I. I. Shapiro, Fourth test of general relativity, *Physical Review Letters* 13, 789 (1964), doi:10.1103/PhysRevLett.13.789.
- [59] S. Coleman and S. L. Glashow, High-energy tests of Lorentz invariance, *Physical Review D* 59, 116008 (1999), doi:10.1103/PhysRevD.59.116008.
- [60] K. Nordtvedt, Probing gravity to the second post-Newtonian order and to one part in  $10^7$  using the spin axis of the Sun, *The Astrophysical Journal* 320, 871 (1987), doi:10.1086/165603.
- [61] T. Jacobson, S. Liberati and D. Mattingly, A strong astrophysical constraint on the violation of special relativity by quantum gravity, *Nature* 424, 1019 (2003), doi:10.1038/nature01882.
- [62] F. W. Stecker, Limiting superluminal electron and neutrino velocities using the 2010 Crab Nebula flare and the IceCube PeV neutrino events, *Astroparticle Physics* 56, 16 (2014), doi:10.1016/j.astropartphys.2014.02.007.
- [63] The Event Horizon Telescope Collaboration, First Sagittarius A\* Event Horizon Telescope results. VI. Testing the black hole metric, *The Astrophysical Journal Letters* 930, L17 (2022), doi:10.3847/2041-8213/ac6756.
- [64] The Event Horizon Telescope Collaboration, First M87 Event Horizon Telescope results. VI. The shadow and mass of the central black hole, *The Astrophysical Journal Letters* 875, L6 (2019), doi:10.3847/2041-8213/ab1141.
- [65] S. Hossenfelder and T. Palmer, Rethinking superdeterminism, *Frontiers in Physics* 8, 139 (2020), doi:10.3389/fphy.2020.00139.
- [66] Z. Cao et al. (LHAASO Collaboration), Ultrahigh-energy photons up to 1.4 petaelectronvolts from 12 gamma-ray Galactic sources, *Nature* 594, 33 (2021), doi:10.1038/s41586-021-03498-z.
- [67] V. A. Acciari et al. (MAGIC Collaboration), Bounds on Lorentz invariance violation from MAGIC observation of GRB 190114C, *Physical Review Letters* 125, 021301 (2020), doi:10.1103/PhysRevLett.125.021301.
- [68] M. Zych, F. Costa, I. Pikovski and C. Brukner, Quantum interferometric visibility as a witness of general relativistic proper time, *Nature Communications* 2, 505 (2011), doi:10.1038/ncomms1498.
- [69] R. C. Tolman, P. Ehrenfest and B. Podolsky, On the gravitational field produced by light, *Physical Review* 37, 602 (1931), doi:10.1103/PhysRev.37.602.
- [70] J. C. Bergquist, R. G. Hulet, W. M. Itano and D. J. Wineland, Observation of quantum jumps in a single atom, *Physical Review Letters* 57, 1699 (1986), doi:10.1103/PhysRevLett.57.1699.
- [71] R. J. Cook and H. J. Kimble, Possibility of direct observation of quantum jumps, *Physical Review Letters* 54, 1023 (1985), doi:10.1103/PhysRevLett.54.1023.
- [72] W. M. Itano, D. J. Heinzen, J. J. Bollinger and D. J. Wineland, Quantum Zeno effect, *Physical Review A* 41, 2295 (1990), doi:10.1103/PhysRevA.41.2295.
- [73] S. W. Hawking, Particle creation by black holes, *Communications in Mathematical Physics* 43, 199 (1975), doi:10.1007/BF02345020.
- [74] C. W. F. Everitt et al., Gravity Probe B: final results of a space experiment to test general relativity, *Physical Review Letters* 106, 221101 (2011), doi:10.1103/PhysRevLett.106.221101.
- [75] L. Shao and N. Wex, New tests of local Lorentz invariance of gravity with small-eccentricity binary pulsars, *Classical and Quantum Gravity* 29, 215018 (2012), doi:10.1088/0264-9381/29/21/215018.
- [76] D. M. Greenberger, M. A. Horne and A. Zeilinger, Going beyond Bell's theorem, in *Bell's Theorem, Quantum Theory and Conceptions of the Universe*, ed. M. Kafatos (Kluwer, Dordrecht, 1989), 69, doi:10.1007/978-94-017-0849-4\_10; N. D. Mermin, Extreme quantum entanglement in a superposition of macroscopically distinct states, *Physical Review Letters* 65, 1838 (1990), doi:10.1103/PhysRevLett.65.1838.
- [77] B. S. Cirel'son, Quantum generalizations of Bell's inequality, *Letters in Mathematical Physics* 4, 93 (1980), doi:10.1007/BF00417500.
- [78] F. J. Hasert et al., Observation of neutrino-like interactions without muon or electron in the Gargamelle neutrino experiment, *Physics Letters B* 46, 138 (1973), doi:10.1016/0370-2693(73)90499-1.
- [79] The ALEPH, DELPHI, L3, OPAL and SLD Collaborations, Precision electroweak measurements on the Z resonance, *Physics Reports* 427, 257 (2006), doi:10.1016/j.physrep.2005.12.006.
- [80] C. S. Wu, E. Ambler, R. W. Hayward, D. D. Hoppes and R. P. Hudson, Experimental test of parity conservation in beta decay, *Physical Review* 105, 1413 (1957), doi:10.1103/PhysRev.105.1413.